2.4 Expectation of a Random Variable

2.4.1 The Discrete Case

If X is a discrete random variable having a probability mass function $p(x)$, then the *expected value* of X is defined by

$$E[X] = \sum_{x:p(x)>0} xp(x)$$

In other words, the expected value of X is a weighted average of the possible values that X can take on, each value being weighted by the probability that X assumes that value. For example, if the probability mass function of X is given by

$$p(1) = \tfrac{1}{2} = p(2)$$

then

$$E[X] = 1\left(\tfrac{1}{2}\right) + 2\left(\tfrac{1}{2}\right) = \tfrac{3}{2}$$

is just an ordinary average of the two possible values 1 and 2 that X can assume. On the other hand, if

$$p(1) = \tfrac{1}{3}, \qquad p(2) = \tfrac{2}{3}$$

then

$$E[X] = 1\left(\tfrac{1}{3}\right) + 2\left(\tfrac{2}{3}\right) = \tfrac{5}{3}$$

is a weighted average of the two possible values 1 and 2 where the value 2 is given twice as much weight as the value 1 since $p(2) = 2p(1)$.

Example 2.15. Find $E[X]$ where X is the outcome when we roll a fair die.

Solution: Since $p(1) = p(2) = p(3) = p(4) = p(5) = p(6) = \frac{1}{6}$, we obtain

$$E[X] = 1\left(\tfrac{1}{6}\right) + 2\left(\tfrac{1}{6}\right) + 3\left(\tfrac{1}{6}\right) + 4\left(\tfrac{1}{6}\right) + 5\left(\tfrac{1}{6}\right) + 6\left(\tfrac{1}{6}\right) = \tfrac{7}{2}$$ ■

Example 2.16 (Expectation of a Bernoulli Random Variable)**.** Calculate $E[X]$ when X is a Bernoulli random variable with parameter p.

Solution: Since $p(0) = 1 - p$, $p(1) = p$, we have

$$E[X] = 0(1 - p) + 1(p) = p$$

Thus, the expected number of successes in a single trial is just the probability that the trial will be a success. ■

Example 2.17 (Expectation of a Binomial Random Variable)**.** Calculate $E[X]$ when X is binomially distributed with parameters n and p.

Solution:

$$\begin{aligned}
E[X] &= \sum_{i=0}^{n} ip(i) \\
&= \sum_{i=0}^{n} i \binom{n}{i} p^i (1-p)^{n-i} \\
&= \sum_{i=1}^{n} \frac{in!}{(n-i)!\,i!} p^i (1-p)^{n-i} \\
&= \sum_{i=1}^{n} \frac{n!}{(n-i)!(i-1)!} p^i (1-p)^{n-i} \\
&= np \sum_{i=1}^{n} \frac{(n-1)!}{(n-i)!(i-1)!} p^{i-1} (1-p)^{n-i} \\
&= np \sum_{k=0}^{n-1} \binom{n-1}{k} p^k (1-p)^{n-1-k} \\
&= np[p + (1-p)]^{n-1} \\
&= np
\end{aligned}$$

where the third from the last equality follows by letting $k = i - 1$. Thus, the expected number of successes in n independent trials is n multiplied by the probability that a trial results in a success. ■

Example 2.18 (Expectation of a Geometric Random Variable)**.** Calculate the expectation of a geometric random variable having parameter p.

Solution: By Eq. (2.4), we have

$$\begin{aligned}
E[X] &= \sum_{n=1}^{\infty} np(1-p)^{n-1} \\
&= p \sum_{n=1}^{\infty} nq^{n-1}
\end{aligned}$$

where $q = 1 - p$,

$$E[X] = p \sum_{n=1}^{\infty} \frac{d}{dq}(q^n)$$

$$
\begin{aligned}
&= p\frac{d}{dq}\left(\sum_{n=1}^{\infty} q^n\right) \\
&= p\frac{d}{dq}\left(\frac{q}{1-q}\right) \\
&= \frac{p}{(1-q)^2} \\
&= \frac{1}{p}
\end{aligned}
$$

In words, the expected number of independent trials we need to perform until we attain our first success equals the reciprocal of the probability that any one trial results in a success. ■

Example 2.19 (Expectation of a Poisson Random Variable)**.** Calculate $E[X]$ if X is a Poisson random variable with parameter λ.

Solution: From Eq. (2.5), we have

$$
\begin{aligned}
E[X] &= \sum_{i=0}^{\infty} \frac{i e^{-\lambda}\lambda^i}{i!} \\
&= \sum_{i=1}^{\infty} \frac{e^{-\lambda}\lambda^i}{(i-1)!} \\
&= \lambda e^{-\lambda} \sum_{i=1}^{\infty} \frac{\lambda^{i-1}}{(i-1)!} \\
&= \lambda e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} \\
&= \lambda e^{-\lambda} e^{\lambda} \\
&= \lambda
\end{aligned}
$$

where we have used the identity $\sum_{k=0}^{\infty} \lambda^k/k! = e^{\lambda}$. ■

2.4.2 The Continuous Case

We may also define the expected value of a continuous random variable. This is done as follows. If X is a continuous random variable having a probability density function $f(x)$, then the expected value of X is defined by

$$
E[X] = \int_{-\infty}^{\infty} x f(x)\, dx
$$

Example 2.20 (Expectation of a Uniform Random Variable)**.** Calculate the expectation of a random variable uniformly distributed over (α, β).

Solution: From Eq. (2.8) we have

$$\begin{aligned} E[X] &= \int_{\alpha}^{\beta} \frac{x}{\beta - \alpha}\, dx \\ &= \frac{\beta^2 - \alpha^2}{2(\beta - \alpha)} \\ &= \frac{\beta + \alpha}{2} \end{aligned}$$

In other words, the expected value of a random variable uniformly distributed over the interval (α, β) is just the midpoint of the interval. ■

Example 2.21 (Expectation of an Exponential Random Variable)**.** Let X be exponentially distributed with parameter λ. Calculate $E[X]$.

Solution:

$$E[X] = \int_0^{\infty} x\lambda e^{-\lambda x}\, dx$$

Integrating by parts ($dv = \lambda e^{-\lambda x}\, dx, u = x$) yields

$$\begin{aligned} E[X] &= -xe^{-\lambda x}\Big|_0^{\infty} + \int_0^{\infty} e^{-\lambda x}\, dx \\ &= 0 - \frac{e^{-\lambda x}}{\lambda}\Big|_0^{\infty} \\ &= \frac{1}{\lambda} \end{aligned}$$

■

Example 2.22 (Expectation of a Normal Random Variable)**.** Calculate $E[X]$ when X is normally distributed with parameters μ and σ^2.

Solution:

$$E[X] = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} xe^{-(x-\mu)^2/2\sigma^2}\, dx$$

Writing x as $(x - \mu) + \mu$ yields

$$E[X] = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} (x - \mu)e^{-(x-\mu)^2/2\sigma^2}\, dx + \mu \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-(x-\mu)^2/2\sigma^2}\, dx$$

Letting $y = x - \mu$ leads to

$$E[X] = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} ye^{-y^2/2\sigma^2} dy + \mu \int_{-\infty}^{\infty} f(x)\, dx$$

where $f(x)$ is the normal density. By symmetry, the first integral must be 0, and so

$$E[X] = \mu \int_{-\infty}^{\infty} f(x)\, dx = \mu$$

■

2.4.3 Expectation of a Function of a Random Variable

Suppose now that we are given a random variable X and its probability distribution (that is, its probability mass function in the discrete case or its probability density function in the continuous case). Suppose also that we are interested in calculating not the expected value of X, but the expected value of some function of X, say, $g(X)$. How do we go about doing this? One way is as follows. Since $g(X)$ is itself a random variable, it must have a probability distribution, which should be computable from a knowledge of the distribution of X. Once we have obtained the distribution of $g(X)$, we can then compute $E[g(X)]$ by the definition of the expectation.

Example 2.23. Suppose X has the following probability mass function:

$$p(0) = 0.2, \qquad p(1) = 0.5, \qquad p(2) = 0.3$$

Calculate $E[X^2]$.

Solution: Letting $Y = X^2$, we have that Y is a random variable that can take on one of the values $0^2, 1^2, 2^2$ with respective probabilities

$$\begin{aligned} p_Y(0) &= P\{Y = 0^2\} = 0.2, \\ p_Y(1) &= P\{Y = 1^2\} = 0.5, \\ p_Y(4) &= P\{Y = 2^2\} = 0.3 \end{aligned}$$

Hence,

$$E[X^2] = E[Y] = 0(0.2) + 1(0.5) + 4(0.3) = 1.7$$

Note that

$$1.7 = E[X^2] \neq (E[X])^2 = 1.21$$ ■

Example 2.24. Let X be uniformly distributed over $(0, 1)$. Calculate $E[X^3]$.

Solution: Letting $Y = X^3$, we calculate the distribution of Y as follows. For $0 \leq a \leq 1$,

$$\begin{aligned} F_Y(a) &= P\{Y \leq a\} \\ &= P\{X^3 \leq a\} \\ &= P\{X \leq a^{1/3}\} \\ &= a^{1/3} \end{aligned}$$

where the last equality follows since X is uniformly distributed over $(0, 1)$. By differentiating $F_Y(a)$, we obtain the density of Y, namely,

$$f_Y(a) = \tfrac{1}{3}a^{-2/3}, \qquad 0 \leq a \leq 1$$

Hence,

$$\begin{aligned} E[X^3] = E[Y] &= \int_{-\infty}^{\infty} a f_Y(a)\, da \\ &= \int_0^1 a \tfrac{1}{3} a^{-2/3}\, da \\ &= \tfrac{1}{3} \int_0^1 a^{1/3}\, da \\ &= \tfrac{1}{3}\tfrac{3}{4} a^{4/3} \Big|_0^1 \\ &= \tfrac{1}{4} \end{aligned}$$

■

While the foregoing procedure will, in theory, always enable us to compute the expectation of any function of X from a knowledge of the distribution of X, there is, fortunately, an easier way to do this. The following proposition shows how we can calculate the expectation of $g(X)$ without first determining its distribution.

Proposition 2.1. **(a)** *If X is a discrete random variable with probability mass function $p(x)$, then for any real-valued function g,*

$$E[g(X)] = \sum_{x:p(x)>0} g(x)p(x)$$

(b) *If X is a continuous random variable with probability density function $f(x)$, then for any real-valued function g,*

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x)\, dx$$

■

Example 2.25. Applying the proposition to Example 2.23 yields

$$E[X^2] = 0^2(0.2) + (1^2)(0.5) + (2^2)(0.3) = 1.7$$

which, of course, checks with the result derived in Example 2.23.

Applying the proposition to Example 2.24 yields

$$\begin{aligned} E[X^3] &= \int_0^1 x^3\, dx \qquad (\text{since } f(x) = 1, 0 < x < 1) \\ &= \tfrac{1}{4} \end{aligned}$$

■

A simple corollary of Proposition 2.1 is the following.

Corollary 2.2. *If a and b are constants, then*

$$E[aX + b] = aE[X] + b$$

Proof. In the discrete case,

$$\begin{aligned} E[aX+b] &= \sum_{x:p(x)>0} (ax+b)p(x) \\ &= a \sum_{x:p(x)>0} xp(x) + b \sum_{x:p(x)>0} p(x) \\ &= aE[X] + b \end{aligned}$$

In the continuous case,

$$\begin{aligned} E[aX+b] &= \int_{-\infty}^{\infty} (ax+b) f(x)\, dx \\ &= a \int_{-\infty}^{\infty} xf(x)\, dx + b \int_{-\infty}^{\infty} f(x)\, dx \\ &= aE[X] + b \end{aligned}$$

■

The expected value of a random variable X, $E[X]$, is also referred to as the *mean* or the first *moment* of X. The quantity $E[X^n]$, $n \geq 1$, is called the nth moment of X. By Proposition 2.1, we note that

$$E[X^n] = \begin{cases} \displaystyle\sum_{x:p(x)>0} x^n p(x), & \text{if } X \text{ is discrete} \\ \displaystyle\int_{-\infty}^{\infty} x^n f(x)\, dx, & \text{if } X \text{ is continuous} \end{cases}$$

Another quantity of interest is the variance of a random variable X, denoted by $\mathrm{Var}(X)$, which is defined by

$$\mathrm{Var}(X) = E\Big[(X - E[X])^2\Big]$$

Thus, the variance of X measures the expected square of the deviation of X from its expected value.

Example 2.26 (Variance of the Normal Random Variable)**.** Let X be normally distributed with parameters μ and σ^2. Find $\mathrm{Var}(X)$.

Solution: Recalling (see Example 2.22) that $E[X] = \mu$, we have that

$$\begin{aligned} \mathrm{Var}(X) &= E[(X-\mu)^2] \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} (x-\mu)^2 e^{-(x-\mu)^2/2\sigma^2}\, dx \end{aligned}$$

Substituting $y = (x-\mu)/\sigma$ yields

$$\mathrm{Var}(X) = \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y^2 e^{-y^2/2}\, dy$$

Integrating by parts ($u = y, dv = ye^{-y^2/2}dy$) gives

$$\begin{aligned}\text{Var}(X) &= \frac{\sigma^2}{\sqrt{2\pi}}\left(-ye^{-y^2/2}\Big|_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-y^2/2}\,dy\right)\\ &= \frac{\sigma^2}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-y^2/2}\,dy\\ &= \sigma^2\end{aligned}$$

Another derivation of $\text{Var}(X)$ will be given in Example 2.42. ■

Suppose that X is continuous with density f, and let $E[X] = \mu$. Then,

$$\begin{aligned}\text{Var}(X) &= E[(X-\mu)^2]\\ &= E[X^2 - 2\mu X + \mu^2]\\ &= \int_{-\infty}^{\infty}(x^2 - 2\mu x + \mu^2)f(x)\,dx\\ &= \int_{-\infty}^{\infty} x^2 f(x)\,dx - 2\mu\int_{-\infty}^{\infty} xf(x)\,dx + \mu^2\int_{-\infty}^{\infty} f(x)\,dx\\ &= E[X^2] - 2\mu\mu + \mu^2\\ &= E[X^2] - \mu^2\end{aligned}$$

A similar proof holds in the discrete case, and so we obtain the useful identity

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Example 2.27. Calculate $\text{Var}(X)$ when X represents the outcome when a fair die is rolled.

Solution: As previously noted in Example 2.15, $E[X] = \frac{7}{2}$. Also,

$$E[X^2] = 1\left(\tfrac{1}{6}\right) + 2^2\left(\tfrac{1}{6}\right) + 3^2\left(\tfrac{1}{6}\right) + 4^2\left(\tfrac{1}{6}\right) + 5^2\left(\tfrac{1}{6}\right) + 6^2\left(\tfrac{1}{6}\right) = \left(\tfrac{1}{6}\right)(91)$$

Hence,

$$\text{Var}(X) = \tfrac{91}{6} - \left(\tfrac{7}{2}\right)^2 = \tfrac{35}{12}$$ ■

2.5 Jointly Distributed Random Variables

2.5.1 Joint Distribution Functions

Thus far, we have concerned ourselves with the probability distribution of a single random variable. However, we are often interested in probability statements concerning two or more random variables. To deal with such probabilities, we define, for any

two random variables X and Y, the *joint cumulative probability distribution function* of X and Y by

$$F(a, b) = P\{X \leq a, Y \leq b\}, \qquad -\infty < a, b < \infty$$

The distribution of X can be obtained from the joint distribution of X and Y as follows:

$$\begin{aligned} F_X(a) &= P\{X \leq a\} \\ &= P\{X \leq a, Y < \infty\} \\ &= F(a, \infty) \end{aligned}$$

Similarly, the cumulative distribution function of Y is given by

$$F_Y(b) = P\{Y \leq b\} = F(\infty, b)$$

In the case where X and Y are both discrete random variables, it is convenient to define the *joint probability mass function* of X and Y by

$$p(x, y) = P\{X = x, Y = y\}$$

The probability mass function of X may be obtained from $p(x, y)$ by

$$p_X(x) = \sum_{y:p(x,y)>0} p(x, y)$$

Similarly,

$$p_Y(y) = \sum_{x:p(x,y)>0} p(x, y)$$

We say that X and Y are *jointly continuous* if there exists a function $f(x, y)$, defined for all real x and y, having the property that for all sets A and B of real numbers

$$P\{X \in A, Y \in B\} = \int_B \int_A f(x, y)\, dx\, dy$$

The function $f(x, y)$ is called the *joint probability density function* of X and Y. The probability density of X can be obtained from a knowledge of $f(x, y)$ by the following reasoning:

$$\begin{aligned} P\{X \in A\} &= P\{X \in A, Y \in (-\infty, \infty)\} \\ &= \int_{-\infty}^{\infty} \int_A f(x, y)\, dx\, dy \\ &= \int_A f_X(x)\, dx \end{aligned}$$

where

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)\, dy$$

is thus the probability density function of X. Similarly, the probability density function of Y is given by

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y)\, dx$$

Because

$$F(a, b) = P(X \le a, Y \le b) = \int_{-\infty}^{a} \int_{-\infty}^{b} f(x, y)\, dy\, dx$$

differentiation yields

$$\frac{d^2}{da\, db} F(a, b) = f(a, b)$$

Thus, as in the single variable case, differentiating the probability distribution function gives the probability density function.

A variation of Proposition 2.1 states that if X and Y are random variables and g is a function of two variables, then

$$\begin{aligned} E[g(X, Y)] &= \sum_y \sum_x g(x, y) p(x, y) && \text{in the discrete case} \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y)\, dx\, dy && \text{in the continuous case} \end{aligned}$$

For example, if $g(X, Y) = X + Y$, then, in the continuous case,

$$\begin{aligned} E[X + Y] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x + y) f(x, y)\, dx\, dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y)\, dx\, dy + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y)\, dx\, dy \\ &= E[X] + E[Y] \end{aligned}$$

where the first integral is evaluated by using the variation of Proposition 2.1 with $g(x, y) = x$, and the second with $g(x, y) = y$.

The same result holds in the discrete case and, combined with the corollary in Section 2.4.3, yields that for any constants a, b

$$E[aX + bY] = aE[X] + bE[Y] \tag{2.10}$$

Joint probability distributions may also be defined for n random variables. The details are exactly the same as when $n = 2$ and are left as an exercise. The corresponding

result to Eq. (2.10) states that if $X_1, X_2, \ldots, X_n$ are n random variables, then for any n constants $a_1, a_2, \ldots, a_n$,

$$E[a_1X_1 + a_2X_2 + \cdots + a_nX_n] = a_1E[X_1] + a_2E[X_2] + \cdots + a_nE[X_n] \quad (2.11)$$

Example 2.28. Calculate the expected sum obtained when three fair dice are rolled.

Solution: Let X denote the sum obtained. Then $X = X_1 + X_2 + X_3$ where X_i represents the value of the ith die. Thus,

$$E[X] = E[X_1] + E[X_2] + E[X_3] = 3\left(\frac{7}{2}\right) = \frac{21}{2}$$ ■

Example 2.29. As another example of the usefulness of Eq. (2.11), let us use it to obtain the expectation of a binomial random variable having parameters n and p. Recalling that such a random variable X represents the number of successes in n trials when each trial has probability p of being a success, we have

$$X = X_1 + X_2 + \cdots + X_n$$

where

$$X_i = \begin{cases} 1, & \text{if the } i\text{th trial is a success} \\ 0, & \text{if the } i\text{th trial is a failure} \end{cases}$$

Hence, X_i is a Bernoulli random variable having expectation $E[X_i] = 1(p) + 0(1 - p) = p$. Thus,

$$E[X] = E[X_1] + E[X_2] + \cdots + E[X_n] = np$$

This derivation should be compared with the one presented in Example 2.17. ■

Example 2.30. At a party N men throw their hats into the center of a room. The hats are mixed up and each man randomly selects one. Find the expected number of men who select their own hats.

Solution: Letting X denote the number of men that select their own hats, we can best compute $E[X]$ by noting that

$$X = X_1 + X_2 + \cdots + X_N$$

where

$$X_i = \begin{cases} 1, & \text{if the } i\text{th man selects his own hat} \\ 0, & \text{otherwise} \end{cases}$$

Now, because the ith man is equally likely to select any of the N hats, it follows that

$$P\{X_i = 1\} = P\{i\text{th man selects his own hat}\} = \frac{1}{N}$$

and so

$$E[X_i] = 1P\{X_i = 1\} + 0P\{X_i = 0\} = \frac{1}{N}$$

Hence, from Eq. (2.11) we obtain

$$E[X] = E[X_1] + \cdots + E[X_N] = \left(\frac{1}{N}\right) N = 1$$

Hence, no matter how many people are at the party, on the average exactly one of the men will select his own hat. ■

Example 2.31. Suppose there are 25 different types of coupons and suppose that each time one obtains a coupon, it is equally likely to be any one of the 25 types. Compute the expected number of different types that are contained in a set of 10 coupons.

Solution: Let X denote the number of different types in the set of 10 coupons. We compute $E[X]$ by using the representation

$$X = X_1 + \cdots + X_{25}$$

where

$$X_i = \begin{cases} 1, & \text{if at least one type } i \text{ coupon is in the set of 10} \\ 0, & \text{otherwise} \end{cases}$$

Now,

$$\begin{aligned} E[X_i] &= P\{X_i = 1\} \\ &= P\{\text{at least one type } i \text{ coupon is in the set of 10}\} \\ &= 1 - P\{\text{no type } i \text{ coupons are in the set of 10}\} \\ &= 1 - \left(\tfrac{24}{25}\right)^{10} \end{aligned}$$

when the last equality follows since each of the 10 coupons will (independently) not be a type i with probability $\frac{24}{25}$. Hence,

$$E[X] = E[X_1] + \cdots + E[X_{25}] = 25\left[1 - \left(\tfrac{24}{25}\right)^{10}\right]$$ ■

Example 2.32. Let $R_1, \ldots, R_{n+m}$ be a random permutation of $1, \ldots, n+m$. (That is, $R_1, \ldots, R_{n+m}$ is equally likely to be any of the $(n+m)!$ permutations of $1, \ldots, n+m$.) For a given $i \le n$, let X be the ith smallest of the values $R_1, \ldots, R_n$. Find $E[X]$.

Solution: If we let N be the number of the values $R_{n+1}, \ldots, R_{n+m}$ that are smaller than X, then X is the $(i+N)$th smallest of all the values $R_1, \ldots, R_{n+m}$. Because $R_1, \ldots, R_{n+m}$ consists of all numbers $1, \ldots, n+m$, it follows that $X = i + N$. Consequently,

$$E[X] = i + E[N]$$

To compute $E[N]$, for $k = 1, \ldots, m$ let I_{n+k} equal 1 if $R_{n+k} < X$ and let it equal 0 otherwise. Using that

$$N = \sum_{k=1}^{m} I_{n+k}$$

we obtain that

$$E[X] = i + \sum_{k=1}^{m} E[I_{n+k}]$$

Now,

$$\begin{aligned} E[I_{n+k}] &= P(R_{n+k} < X) \\ &= P(R_{n+k} < i\text{th smallest of } R_1, \ldots, R_n) \\ &= P(R_{n+k} \text{ is one of the } i \text{ smallest of the values } R_1, \ldots, R_n, R_{n+k}) \\ &= \frac{i}{n+1} \end{aligned}$$

where the final equality used that R_{n+k} is equally likely to be either the smallest, the second smallest, ..., or the $(n+1)$st smallest of the values $R_1, \ldots, R_n, R_{n+k}$. Hence,

$$E[X] = i + m\frac{i}{n+1}$$ ■

2.5.2 Independent Random Variables

The random variables X and Y are said to be *independent* if, for all a, b,

$$P\{X \le a, Y \le b\} = P\{X \le a\}P\{Y \le b\} \tag{2.12}$$

In other words, X and Y are independent if, for all a and b, the events $E_a = \{X \le a\}$ and $F_b = \{Y \le b\}$ are independent.

In terms of the joint distribution function F of X and Y, we have that X and Y are independent if

$$F(a, b) = F_X(a)F_Y(b) \qquad \text{for all } a, b$$

When X and Y are discrete, the condition of independence reduces to

$$p(x, y) = p_X(x)p_Y(y) \tag{2.13}$$

while if X and Y are jointly continuous, independence reduces to

$$f(x, y) = f_X(x)f_Y(y) \tag{2.14}$$

To prove this statement, consider first the discrete version, and suppose that the joint probability mass function $p(x, y)$ satisfies Eq. (2.13). Then

$$\begin{aligned} P\{X \le a, Y \le b\} &= \sum_{y \le b} \sum_{x \le a} p(x, y) \\ &= \sum_{y \le b} \sum_{x \le a} p_X(x) p_Y(y) \\ &= \sum_{y \le b} p_Y(y) \sum_{x \le a} p_X(x) \\ &= P\{Y \le b\} P\{X \le a\} \end{aligned}$$

and so X and Y are independent. That Eq. (2.14) implies independence in the continuous case is proven in the same manner and is left as an exercise.

An important result concerning independence is the following.

Proposition 2.3. *If X and Y are independent, then for any functions h and g*

$$E[g(X)h(Y)] = E[g(X)]E[h(Y)]$$

Proof. Suppose that X and Y are jointly continuous. Then

$$\begin{aligned} E[g(X)h(Y)] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x)h(y) f(x, y)\, dx\, dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x)h(y) f_X(x) f_Y(y)\, dx\, dy \\ &= \int_{-\infty}^{\infty} h(y) f_Y(y)\, dy \int_{-\infty}^{\infty} g(x) f_X(x)\, dx \\ &= E[h(Y)]E[g(X)] \end{aligned}$$

The proof in the discrete case is similar. ■

2.5.3 Covariance and Variance of Sums of Random Variables

The covariance of any two random variables X and Y, denoted by $\mathrm{Cov}(X, Y)$, is defined by

$$\begin{aligned} \mathrm{Cov}(X, Y) &= E[(X - E[X])(Y - E[Y])] \\ &= E[XY - YE[X] - XE[Y] + E[X]E[Y]] \\ &= E[XY] - E[Y]E[X] - E[X]E[Y] + E[X]E[Y] \\ &= E[XY] - E[X]E[Y] \end{aligned}$$

Note that if X and Y are independent, then by Proposition 2.3 it follows that $\mathrm{Cov}(X, Y) = 0$.

Let us consider now the special case where X and Y are indicator variables for whether or not the events A and B occur. That is, for events A and B, define

$$X = \begin{cases} 1, & \text{if } A \text{ occurs} \\ 0, & \text{otherwise,} \end{cases} \qquad Y = \begin{cases} 1, & \text{if } B \text{ occurs} \\ 0, & \text{otherwise} \end{cases}$$

Then,

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

and, because XY will equal 1 or 0 depending on whether or not both X and Y equal 1, we see that

$$\text{Cov}(X, Y) = P\{X = 1, Y = 1\} - P\{X = 1\}P\{Y = 1\}$$

From this we see that

$$\begin{aligned} \text{Cov}(X, Y) > 0 &\Leftrightarrow P\{X = 1, Y = 1\} > P\{X = 1\}P\{Y = 1\} \\ &\Leftrightarrow \frac{P\{X = 1, Y = 1\}}{P\{X = 1\}} > P\{Y = 1\} \\ &\Leftrightarrow P\{Y = 1|X = 1\} > P\{Y = 1\} \end{aligned}$$

That is, the covariance of X and Y is positive if the outcome $X = 1$ makes it more likely that $Y = 1$ (which, as is easily seen by symmetry, also implies the reverse).

In general it can be shown that a positive value of $\text{Cov}(X, Y)$ is an indication that Y tends to increase as X does, whereas a negative value indicates that Y tends to decrease as X increases.

Example 2.33. The joint density function of X, Y is

$$f(x, y) = \frac{1}{y} e^{-(y+x/y)}, \quad 0 < x, y < \infty$$

(a) Verify that the preceding is a joint density function.
(b) Find Cov (X, Y).

Solution: To show that $f(x, y)$ is a joint density function we need to show it is nonnegative, which is immediate, and that $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dy\, dx = 1$. We prove the latter as follows:

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dy\, dx &= \int_0^{\infty} \int_0^{\infty} \frac{1}{y} e^{-(y+x/y)} dy\, dx \\ &= \int_0^{\infty} e^{-y} \int_0^{\infty} \frac{1}{y} e^{-x/y} dx\, dy \\ &= \int_0^{\infty} e^{-y} dy \\ &= 1 \end{aligned}$$

To obtain $\text{Cov}(X, Y)$, note that the density function of Y is

$$f_Y(y) = e^{-y} \int_0^\infty \frac{1}{y} e^{-x/y} dx = e^{-y}$$

Thus, Y is an exponential random variable with parameter 1, showing (see Example 2.21) that

$$E[Y] = 1$$

We compute $E[X]$ and $E[XY]$ as follows:

$$\begin{aligned} E[X] &= \int_{-\infty}^\infty \int_{-\infty}^\infty x f(x, y)\, dy\, dx \\ &= \int_0^\infty e^{-y} \int_0^\infty \frac{x}{y} e^{-x/y} dx\, dy \end{aligned}$$

Now, $\int_0^\infty \frac{x}{y} e^{-x/y} dx$ is the expected value of an exponential random variable with parameter $1/y$, and thus is equal to y. Consequently,

$$E[X] = \int_0^\infty y e^{-y} dy = 1$$

Also

$$\begin{aligned} E[XY] &= \int_{-\infty}^\infty \int_{-\infty}^\infty xy\, f(x, y)\, dy\, dx \\ &= \int_0^\infty y e^{-y} \int_0^\infty \frac{x}{y} e^{-x/y} dx\, dy \\ &= \int_0^\infty y^2 e^{-y} dy \end{aligned}$$

Integration by parts ($dv = e^{-y} dy, u = y^2$) gives

$$E[XY] = \int_0^\infty y^2 e^{-y} dy = -y^2 e^{-y}\big|_0^\infty + \int_0^\infty 2y e^{-y} dy = 2E[Y] = 2$$

Consequently,

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 1$$ ■

The following are important properties of covariance.

Properties of Covariance

For any random variables X, Y, Z and constant c,

1. $\text{Cov}(X, X) = \text{Var}(X)$,

2. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$,
3. $\text{Cov}(cX, Y) = c\,\text{Cov}(X, Y)$,
4. $\text{Cov}(X, Y + Z) = \text{Cov}(X, Y) + \text{Cov}(X, Z)$.

Whereas the first three properties are immediate, the final one is easily proven as follows:

$$\begin{aligned}\text{Cov}(X, Y + Z) &= E[X(Y + Z)] - E[X]E[Y + Z] \\ &= E[XY] - E[X]E[Y] + E[XZ] - E[X]E[Z] \\ &= \text{Cov}(X, Y) + \text{Cov}(X, Z)\end{aligned}$$

The fourth property listed easily generalizes to give the following result:

$$\text{Cov}\left(\sum_{i=1}^{n} X_i, \sum_{j=1}^{m} Y_j\right) = \sum_{i=1}^{n}\sum_{j=1}^{m} \text{Cov}(X_i, Y_j) \tag{2.15}$$

A useful expression for the variance of the sum of random variables can be obtained from Eq. (2.15) as follows:

$$\begin{aligned}\text{Var}\left(\sum_{i=1}^{n} X_i\right) &= \text{Cov}\left(\sum_{i=1}^{n} X_i, \sum_{j=1}^{n} X_j\right) \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} \text{Cov}(X_i, X_j) \\ &= \sum_{i=1}^{n} \text{Cov}(X_i, X_i) + \sum_{i=1}^{n}\sum_{j\neq i} \text{Cov}(X_i, X_j) \\ &= \sum_{i=1}^{n} \text{Var}(X_i) + 2\sum_{i=1}^{n}\sum_{j<i} \text{Cov}(X_i, X_j)\end{aligned} \tag{2.16}$$

If $X_i, i = 1, \ldots, n$ are independent random variables, then Eq. (2.16) reduces to

$$\text{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \text{Var}(X_i)$$

Definition 2.1. If $X_1, \ldots, X_n$ are independent and identically distributed, then the random variable $\bar{X} = \sum_{i=1}^{n} X_i/n$ is called the *sample mean*.

The following proposition shows that the covariance between the sample mean and a deviation from that sample mean is zero. It will be needed in Section 2.6.1.

Proposition 2.4. *Suppose that $X_1, \ldots, X_n$ are independent and identically distributed with expected value μ and variance σ^2. Then,*

(a) $E[\bar{X}] = \mu$.

(b) $\text{Var}(\bar{X}) = \sigma^2/n$.
(c) $\text{Cov}(\bar{X}, X_i - \bar{X}) = 0, i = 1, \ldots, n$.

Proof. Parts (a) and (b) are easily established as follows:

$$E[\bar{X}] = \frac{1}{n}\sum_{i=1}^{n} E[X_i] = \mu,$$

$$\text{Var}(\bar{X}) = \left(\frac{1}{n}\right)^2 \text{Var}\left(\sum_{i=1}^{n} X_i\right) = \left(\frac{1}{n}\right)^2 \sum_{i=1}^{n} \text{Var}(X_i) = \frac{\sigma^2}{n}$$

To establish part (c) we reason as follows:

$$\begin{aligned}
\text{Cov}(\bar{X}, X_i - \bar{X}) &= \text{Cov}(\bar{X}, X_i) - \text{Cov}(\bar{X}, \bar{X}) \\
&= \frac{1}{n}\text{Cov}\left(X_i + \sum_{j \neq i} X_j, X_i\right) - \text{Var}(\bar{X}) \\
&= \frac{1}{n}\text{Cov}(X_i, X_i) + \frac{1}{n}\text{Cov}\left(\sum_{j \neq i} X_j, X_i\right) - \frac{\sigma^2}{n} \\
&= \frac{\sigma^2}{n} - \frac{\sigma^2}{n} = 0
\end{aligned}$$

where the next to the last equality used the fact that X_i and $\sum_{j \neq i} X_j$ are independent and thus have covariance 0. ■

Eq. (2.16) is often useful when computing variances.

Example 2.34 (Variance of a Binomial Random Variable)**.** Compute the variance of a binomial random variable X with parameters n and p.

Solution: Since such a random variable represents the number of successes in n independent trials when each trial has a common probability p of being a success, we may write

$$X = X_1 + \cdots + X_n$$

where the X_i are independent Bernoulli random variables such that

$$X_i = \begin{cases} 1, & \text{if the } i\text{th trial is a success} \\ 0, & \text{otherwise} \end{cases}$$

Hence, from Eq. (2.16) we obtain

$$\text{Var}(X) = \text{Var}(X_1) + \cdots + \text{Var}(X_n)$$

But

$$\text{Var}(X_i) = E[X_i^2] - (E[X_i])^2$$

$$= E[X_i] - (E[X_i])^2 \qquad \text{since } X_i^2 = X_i$$
$$= p - p^2$$

and thus

$$\text{Var}(X) = np(1-p)$$ ■

Example 2.35 (Sampling from a Finite Population: The Hypergeometric)**.** Consider a population of N individuals, some of whom are in favor of a certain proposition. In particular suppose that Np of them are in favor and $N - Np$ are opposed, where p is assumed to be unknown. We are interested in estimating p, the fraction of the population that is for the proposition, by randomly choosing and then determining the positions of n members of the population.

In such situations as described in the preceding, it is common to use the fraction of the sampled population that is in favor of the proposition as an estimator of p. Hence, if we let

$$X_i = \begin{cases} 1, & \text{if the } i\text{th person chosen is in favor} \\ 0, & \text{otherwise} \end{cases}$$

then the usual estimator of p is $\sum_{i=1}^{n} X_i/n$. Let us now compute its mean and variance. Now,

$$E\left[\sum_{i=1}^{n} X_i\right] = \sum_{1}^{n} E[X_i]$$
$$= np$$

where the final equality follows since the ith person chosen is equally likely to be any of the N individuals in the population and so has probability Np/N of being in favor.

$$\text{Var}\left(\sum_{1}^{n} X_i\right) = \sum_{1}^{n} \text{Var}(X_i) + 2\sum\sum_{i<j} \text{Cov}(X_i, X_j)$$

Now, since X_i is a Bernoulli random variable with mean p, it follows that

$$\text{Var}(X_i) = p(1-p)$$

Also, for $i \neq j$,

$$\text{Cov}(X_i, X_j) = E[X_i X_j] - E[X_i]E[X_j]$$
$$= P\{X_i = 1, X_j = 1\} - p^2$$
$$= P\{X_i = 1\}P\{X_j = 1 | X_i = 1\} - p^2$$
$$= \frac{Np}{N}\frac{(Np-1)}{N-1} - p^2$$

where the last equality follows since if the ith person to be chosen is in favor, then the jth person chosen is equally likely to be any of the other $N-1$ of which $Np-1$ are in favor. Thus, we see that

$$\begin{aligned}\text{Var}\left(\sum_1^n X_i\right) &= np(1-p) + 2\binom{n}{2}\left[\frac{p(Np-1)}{N-1} - p^2\right] \\ &= np(1-p) - \frac{n(n-1)p(1-p)}{N-1}\end{aligned}$$

and so the mean and variance of our estimator are given by

$$\begin{aligned}E\left[\sum_1^n \frac{X_i}{n}\right] &= p, \\ \text{Var}\left[\sum_1^n \frac{X_i}{n}\right] &= \frac{p(1-p)}{n} - \frac{(n-1)p(1-p)}{n(N-1)}\end{aligned}$$

Some remarks are in order: As the mean of the estimator is the unknown value p, we would like its variance to be as small as possible (why is this?), and we see by the preceding that, as a function of the population size N, the variance increases as N increases. The limiting value, as $N \to \infty$, of the variance is $p(1-p)/n$, which is not surprising since for N large each of the X_i will be (approximately) independent random variables, and thus $\sum_1^n X_i$ will have an (approximately) binomial distribution with parameters n and p.

The random variable $\sum_1^n X_i$ can be thought of as representing the number of white balls obtained when n balls are randomly selected from a population consisting of Np white and $N - Np$ black balls. (Identify a person who favors the proposition with a white ball and one against with a black ball.) Such a random variable is called *hypergeometric* and has a probability mass function given by

$$P\left\{\sum_1^n X_i = k\right\} = \frac{\binom{Np}{k}\binom{N-Np}{n-k}}{\binom{N}{n}} \qquad \blacksquare$$

It is often important to be able to calculate the distribution of $X + Y$ from the distributions of X and Y when X and Y are independent. Suppose first that X and Y are continuous, X having probability density f and Y having probability density g. Then, letting $F_{X+Y}(a)$ be the cumulative distribution function of $X + Y$, we have

$$\begin{aligned}F_{X+Y}(a) &= P\{X+Y \le a\} \\ &= \iint_{x+y\le a} f(x)g(y)\,dx\,dy \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{a-y} f(x)g(y)\,dx\,dy\end{aligned}$$

$$= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{a-y} f(x)\,dx \right) g(y)\,dy$$
$$= \int_{-\infty}^{\infty} F_X(a-y)g(y)\,dy \tag{2.17}$$

The cumulative distribution function F_{X+Y} is called the *convolution* of the distributions F_X and F_Y (the cumulative distribution functions of X and Y, respectively).

By differentiating Eq. (2.17), we obtain that the probability density function $f_{X+Y}(a)$ of $X+Y$ is given by

$$f_{X+Y}(a) = \frac{d}{da} \int_{-\infty}^{\infty} F_X(a-y)g(y)\,dy$$
$$= \int_{-\infty}^{\infty} \frac{d}{da}(F_X(a-y))g(y)\,dy$$
$$= \int_{-\infty}^{\infty} f(a-y)g(y)\,dy \tag{2.18}$$

Example 2.36 (Sum of Two Independent Uniform Random Variables)**.** If X and Y are independent random variables both uniformly distributed on (0, 1), then calculate the probability density of $X+Y$.

Solution: From Eq. (2.18), since

$$f(a) = g(a) = \begin{cases} 1, & 0 < a < 1 \\ 0, & \text{otherwise} \end{cases}$$

we obtain

$$f_{X+Y}(a) = \int_0^1 f(a-y)\,dy$$

For $0 \le a \le 1$, this yields

$$f_{X+Y}(a) = \int_0^a dy = a$$

For $1 < a < 2$, we get

$$f_{X+Y}(a) = \int_{a-1}^1 dy = 2-a$$

Hence,

$$f_{X+Y}(a) = \begin{cases} a, & 0 \le a \le 1 \\ 2-a, & 1 < a < 2 \\ 0, & \text{otherwise} \end{cases}$$

■

Rather than deriving a general expression for the distribution of $X+Y$ in the discrete case, we shall consider an example.

Example 2.37 (Sums of Independent Poisson Random Variables)**.** Let X and Y be independent Poisson random variables with respective means λ_1 and λ_2. Calculate the distribution of $X + Y$.

Solution: Since the event $\{X+Y=n\}$ may be written as the union of the disjoint events $\{X=k, Y=n-k\}, 0 \le k \le n$, we have

$$\begin{aligned}
P\{X+Y=n\} &= \sum_{k=0}^{n} P\{X=k, Y=n-k\} \\
&= \sum_{k=0}^{n} P\{X=k\}P\{Y=n-k\} \\
&= \sum_{k=0}^{n} e^{-\lambda_1}\frac{\lambda_1^k}{k!}e^{-\lambda_2}\frac{\lambda_2^{n-k}}{(n-k)!} \\
&= e^{-(\lambda_1+\lambda_2)}\sum_{k=0}^{n}\frac{\lambda_1^k\lambda_2^{n-k}}{k!(n-k)!} \\
&= \frac{e^{-(\lambda_1+\lambda_2)}}{n!}\sum_{k=0}^{n}\frac{n!}{k!(n-k)!}\lambda_1^k\lambda_2^{n-k} \\
&= \frac{e^{-(\lambda_1+\lambda_2)}}{n!}(\lambda_1+\lambda_2)^n
\end{aligned}$$

In words, $X + Y$ has a Poisson distribution with mean $\lambda_1 + \lambda_2$. ■

The concept of independence may, of course, be defined for more than two random variables. In general, the n random variables $X_1, X_2, \ldots, X_n$ are said to be independent if, for all values $a_1, a_2, \ldots, a_n$,

$$\begin{aligned}
&P\{X_1 \le a_1, X_2 \le a_2, \ldots, X_n \le a_n\} \\
&\quad = P\{X_1 \le a_1\}P\{X_2 \le a_2\}\cdots P\{X_n \le a_n\}
\end{aligned}$$

Example 2.38. Let $X_1, \ldots, X_n$ be independent and identically distributed continuous random variables with probability distribution F and density function $F' = f$. If we let $X_{(i)}$ denote the ith smallest of these random variables, then $X_{(1)}, \ldots, X_{(n)}$ are called the *order statistics*. To obtain the distribution of $X_{(i)}$, note that $X_{(i)}$ will be less than or equal to x if and only if at least i of the n random variables $X_1, \ldots, X_n$ are less than or equal to x. Hence,

$$P\{X_{(i)} \le x\} = \sum_{k=i}^{n}\binom{n}{k}(F(x))^k(1-F(x))^{n-k}$$

Differentiation yields that the density function of $X_{(i)}$ is as follows:

$$f_{X_{(i)}}(x) = f(x)\sum_{k=i}^{n}\binom{n}{k}k(F(x))^{k-1}(1-F(x))^{n-k}$$

$$
\begin{aligned}
&- f(x) \sum_{k=i}^{n} \binom{n}{k} (n-k)(F(x))^k (1-F(x))^{n-k-1} \\
&= f(x) \sum_{k=i}^{n} \frac{n!}{(n-k)!(k-1)!} (F(x))^{k-1} (1-F(x))^{n-k} \\
&\quad - f(x) \sum_{k=i}^{n-1} \frac{n!}{(n-k-1)!k!} (F(x))^k (1-F(x))^{n-k-1} \\
&= f(x) \sum_{k=i}^{n} \frac{n!}{(n-k)!(k-1)!} (F(x))^{k-1} (1-F(x))^{n-k} \\
&\quad - f(x) \sum_{j=i+1}^{n} \frac{n!}{(n-j)!(j-1)!} (F(x))^{j-1} (1-F(x))^{n-j} \\
&= \frac{n!}{(n-i)!(i-1)!} f(x) (F(x))^{i-1} (1-F(x))^{n-i}
\end{aligned}
$$

The preceding density is quite intuitive, since in order for $X_{(i)}$ to equal x, $i-1$ of the n values $X_1, \ldots, X_n$ must be less than x; $n-i$ of them must be greater than x; and one must be equal to x. Now, the probability density that every member of a specified set of $i-1$ of the X_j is less than x, every member of another specified set of $n-i$ is greater than x, and the remaining value is equal to x is $(F(x))^{i-1}(1-F(x))^{n-i} f(x)$. Therefore, since there are $n!/[(i-1)!(n-i)!]$ different partitions of the n random variables into the three groups, we obtain the preceding density function. ■

2.5.4 Joint Probability Distribution of Functions of Random Variables

Let X_1 and X_2 be jointly continuous random variables with joint probability density function $f(x_1, x_2)$. It is sometimes necessary to obtain the joint distribution of the random variables Y_1 and Y_2 that arise as functions of X_1 and X_2. Specifically, suppose that $Y_1 = g_1(X_1, X_2)$ and $Y_2 = g_2(X_1, X_2)$ for some functions g_1 and g_2.

Assume that the functions g_1 and g_2 satisfy the following conditions:

1. The equations $y_1 = g_1(x_1, x_2)$ and $y_2 = g_2(x_1, x_2)$ can be uniquely solved for x_1 and x_2 in terms of y_1 and y_2 with solutions given by, say, $x_1 = h_1(y_1, y_2), x_2 = h_2(y_1, y_2)$.
2. The functions g_1 and g_2 have continuous partial derivatives at all points (x_1, x_2) and are such that the following 2×2 determinant

$$
J(x_1, x_2) = \begin{vmatrix} \dfrac{\partial g_1}{\partial x_1} & \dfrac{\partial g_1}{\partial x_2} \\ \dfrac{\partial g_2}{\partial x_1} & \dfrac{\partial g_2}{\partial x_2} \end{vmatrix} \equiv \frac{\partial g_1}{\partial x_1}\frac{\partial g_2}{\partial x_2} - \frac{\partial g_1}{\partial x_2}\frac{\partial g_2}{\partial x_1} \neq 0
$$

at all points (x_1, x_2).

Under these two conditions it can be shown that the random variables Y_1 and Y_2 are jointly continuous with joint density function given by

$$f_{Y_1,Y_2}(y_1, y_2) = f_{X_1,X_2}(x_1, x_2)|J(x_1, x_2)|^{-1} \tag{2.19}$$

where $x_1 = h_1(y_1, y_2), x_2 = h_2(y_1, y_2)$.

A proof of Eq. (2.19) would proceed along the following lines:

$$P\{Y_1 \le y_1, Y_2 \le y_2\} = \iint\limits_{\substack{(x_1,x_2):\\ g_1(x_1,x_2)\le y_1\\ g_2(x_1,x_2)\le y_2}} f_{X_1,X_2}(x_1, x_2)\, dx_1\, dx_2 \tag{2.20}$$

The joint density function can now be obtained by differentiating Eq. (2.20) with respect to y_1 and y_2. That the result of this differentiation will be equal to the right-hand side of Eq. (2.19) is an exercise in advanced calculus whose proof will not be presented in the present text.

Example 2.39. If X and Y are independent gamma random variables with parameters (α, λ) and (β, λ), respectively, compute the joint density of $U = X + Y$ and $V = X/(X + Y)$.

Solution: The joint density of X and Y is given by

$$\begin{aligned} f_{X,Y}(x, y) &= \frac{\lambda e^{-\lambda x}(\lambda x)^{\alpha-1}}{\Gamma(\alpha)} \frac{\lambda e^{-\lambda y}(\lambda y)^{\beta-1}}{\Gamma(\beta)} \\ &= \frac{\lambda^{\alpha+\beta}}{\Gamma(\alpha)\Gamma(\beta)} e^{-\lambda(x+y)} x^{\alpha-1} y^{\beta-1} \end{aligned}$$

Now, if $g_1(x, y) = x + y$, $g_2(x, y) = x/(x + y)$, then

$$\frac{\partial g_1}{\partial x} = \frac{\partial g_1}{\partial y} = 1, \qquad \frac{\partial g_2}{\partial x} = \frac{y}{(x+y)^2}, \qquad \frac{\partial g_2}{\partial y} = -\frac{x}{(x+y)^2}$$

and so

$$J(x, y) = \begin{vmatrix} 1 & 1 \\ \dfrac{y}{(x+y)^2} & \dfrac{-x}{(x+y)^2} \end{vmatrix} = -\frac{1}{x+y}$$

Finally, because the equations $u = x + y, v = x/(x + y)$ have as their solutions $x = uv, y = u(1 - v)$, we see that

$$\begin{aligned} f_{U,V}(u, v) &= f_{X,Y}[uv, u(1 - v)]u \\ &= \frac{\lambda e^{-\lambda u}(\lambda u)^{\alpha+\beta-1}}{\Gamma(\alpha+\beta)} \frac{v^{\alpha-1}(1-v)^{\beta-1}\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \end{aligned}$$

Hence $X + Y$ and $X/(X + Y)$ are independent, with $X + Y$ having a gamma distribution with parameters $(\alpha + \beta, \lambda)$ and $X/(X + Y)$ having density function

$$f_V(v) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} v^{\alpha-1}(1 - v)^{\beta-1}, \qquad 0 < v < 1$$

This is called the beta density with parameters (α, β).

This result is quite interesting. For suppose there are $n + m$ jobs to be performed, with each (independently) taking an exponential amount of time with rate λ for performance, and suppose that we have two workers to perform these jobs. Worker I will do jobs $1, 2, \ldots, n$, and worker II will do the remaining m jobs. If we let X and Y denote the total working times of workers I and II, respectively, then upon using the preceding result it follows that X and Y will be independent gamma random variables having parameters (n, λ) and (m, λ), respectively. Then the preceding result yields that independently of the working time needed to complete all $n + m$ jobs (that is, of $X + Y$), the proportion of this work that will be performed by worker I has a beta distribution with parameters (n, m). ■

When the joint density function of the n random variables $X_1, X_2, \ldots, X_n$ is given and we want to compute the joint density function of $Y_1, Y_2, \ldots, Y_n$, where

$$Y_1 = g_1(X_1, \ldots, X_n), \qquad Y_2 = g_2(X_1, \ldots, X_n), \qquad \ldots,$$
$$Y_n = g_n(X_1, \ldots, X_n)$$

the approach is the same. Namely, we assume that the functions g_i have continuous partial derivatives and that the Jacobian determinant $J(x_1, \ldots, x_n) \neq 0$ at all points $(x_1, \ldots, x_n)$, where

$$J(x_1, \ldots, x_n) = \begin{vmatrix} \dfrac{\partial g_1}{\partial x_1} & \dfrac{\partial g_1}{\partial x_2} & \cdots & \dfrac{\partial g_1}{\partial x_n} \\ \dfrac{\partial g_2}{\partial x_1} & \dfrac{\partial g_2}{\partial x_2} & \cdots & \dfrac{\partial g_2}{\partial x_n} \\ \dfrac{\partial g_n}{\partial x_1} & \dfrac{\partial g_n}{\partial x_2} & \cdots & \dfrac{\partial g_n}{\partial x_n} \end{vmatrix}$$

Furthermore, we suppose that the equations $y_1 = g_1(x_1, \ldots, x_n)$, $y_2 = g_2(x_1, \ldots, x_n)$, $\ldots, y_n = g_n(x_1, \ldots, x_n)$ have a unique solution, say, $x_1 = h_1(y_1, \ldots, y_n), \ldots, x_n = h_n(y_1, \ldots, y_n)$. Under these assumptions the joint density function of the random variables Y_i is given by

$$f_{Y_1,\ldots,Y_n}(y_1, \ldots, y_n) = f_{X_1,\ldots,X_n}(x_1, \ldots, x_n)|J(x_1, \ldots, x_n)|^{-1}$$

where $x_i = h_i(y_1, \ldots, y_n), i = 1, 2, \ldots, n$.

2.6 Moment Generating Functions

The *moment generating function* $\phi(t)$ of the random variable X is defined for all values t by

$$\begin{aligned}\phi(t) &= E[e^{tX}] \\ &= \begin{cases} \sum_x e^{tx} p(x), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} e^{tx} f(x)\,dx, & \text{if } X \text{ is continuous} \end{cases}\end{aligned}$$

We call $\phi(t)$ the moment generating function because all of the moments of X can be obtained by successively differentiating $\phi(t)$. For example,

$$\begin{aligned}\phi'(t) &= \frac{d}{dt} E[e^{tX}] \\ &= E\left[\frac{d}{dt}(e^{tX})\right] \\ &= E[Xe^{tX}]\end{aligned}$$

Hence,

$$\phi'(0) = E[X]$$

Similarly,

$$\begin{aligned}\phi''(t) &= \frac{d}{dt}\phi'(t) \\ &= \frac{d}{dt} E[Xe^{tX}] \\ &= E\left[\frac{d}{dt}(Xe^{tX})\right] \\ &= E[X^2 e^{tX}]\end{aligned}$$

and so

$$\phi''(0) = E[X^2]$$

In general, the nth derivative of $\phi(t)$ evaluated at $t = 0$ equals $E[X^n]$, that is,

$$\phi^n(0) = E[X^n], \qquad n \geq 1$$

We now compute $\phi(t)$ for some common distributions.

Example 2.40 (The Binomial Distribution with Parameters n and p)**.**

$$\phi(t) = E[e^{tX}]$$

$$
\begin{aligned}
&= \sum_{k=0}^{n} e^{tk} \binom{n}{k} p^k (1-p)^{n-k} \\
&= \sum_{k=0}^{n} \binom{n}{k} (pe^t)^k (1-p)^{n-k} \\
&= (pe^t + 1 - p)^n
\end{aligned}
$$

Hence,

$$\phi'(t) = n(pe^t + 1 - p)^{n-1} pe^t$$

and so

$$E[X] = \phi'(0) = np$$

which checks with the result obtained in Example 2.17. Differentiating a second time yields

$$\phi''(t) = n(n-1)(pe^t + 1 - p)^{n-2}(pe^t)^2 + n(pe^t + 1 - p)^{n-1} pe^t$$

and so

$$E[X^2] = \phi''(0) = n(n-1)p^2 + np$$

Thus, the variance of X is given by

$$
\begin{aligned}
\mathrm{Var}(X) &= E[X^2] - (E[X])^2 \\
&= n(n-1)p^2 + np - n^2 p^2 \\
&= np(1-p)
\end{aligned}
$$

■

Example 2.41 (The Poisson Distribution with Mean λ)**.**

$$
\begin{aligned}
\phi(t) &= E[e^{tX}] \\
&= \sum_{n=0}^{\infty} \frac{e^{tn} e^{-\lambda} \lambda^n}{n!} \\
&= e^{-\lambda} \sum_{n=0}^{\infty} \frac{(\lambda e^t)^n}{n!} \\
&= e^{-\lambda} e^{\lambda e^t} \\
&= \exp\{\lambda(e^t - 1)\}
\end{aligned}
$$

Differentiation yields

$$
\begin{aligned}
&\phi'(t) = \lambda e^t \exp\{\lambda(e^t - 1)\}, \\
&\phi''(t) = (\lambda e^t)^2 \exp\{\lambda(e^t - 1)\} + \lambda e^t \exp\{\lambda(e^t - 1)\}
\end{aligned}
$$

and so

$$\begin{aligned} E[X] &= \phi'(0) = \lambda, \\ E[X^2] &= \phi''(0) = \lambda^2 + \lambda, \\ \mathrm{Var}(X) &= E[X^2] - (E[X])^2 \\ &= \lambda \end{aligned}$$

Thus, both the mean and the variance of the Poisson equal λ. ■

Example 2.42 (The Exponential Distribution with Parameter λ)**.**

$$\begin{aligned} \phi(t) &= E[e^{tX}] \\ &= \int_0^\infty e^{tx} \lambda e^{-\lambda x} dx \\ &= \lambda \int_0^\infty e^{-(\lambda - t)x} dx \\ &= \frac{\lambda}{\lambda - t} \qquad \text{for } t < \lambda \end{aligned}$$

We note by the preceding derivation that, for the exponential distribution, $\phi(t)$ is only defined for values of t less than λ. Differentiation of $\phi(t)$ yields

$$\phi'(t) = \frac{\lambda}{(\lambda - t)^2}, \qquad \phi''(t) = \frac{2\lambda}{(\lambda - t)^3}$$

Hence,

$$E[X] = \phi'(0) = \frac{1}{\lambda}, \qquad E[X^2] = \phi''(0) = \frac{2}{\lambda^2}$$

The variance of X is thus given by

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2 = \frac{1}{\lambda^2}$$ ■

Example 2.43 (The Normal Distribution with Parameters μ and σ^2)**.** The moment generating function of a standard normal random variable Z is obtained as follows.

$$\begin{aligned} E[e^{tZ}] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty e^{tx} e^{-x^2/2} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty e^{-(x^2 - 2tx)/2} dx \\ &= e^{t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty e^{-(x-t)^2/2} dx \\ &= e^{t^2/2} \end{aligned}$$

Table 2.1

Discrete probability distribution	Probability mass function, $p(x)$	Moment generating function, $\phi(t)$	Mean	Variance
Binomial with parameters n, p, $0 \le p \le 1$	$\binom{n}{x} p^x (1-p)^{n-x}$, $x = 0, 1, \ldots, n$	$(pe^t + (1-p))^n$	np	$np(1-p)$
Poisson with parameter $\lambda > 0$	$e^{-\lambda}\frac{\lambda^x}{x!}$, $x = 0, 1, 2, \ldots$	$\exp\{\lambda(e^t - 1)\}$	λ	λ
Geometric with parameter $0 \le p \le 1$	$p(1-p)^{x-1}$, $x = 1, 2, \ldots$	$\frac{pe^t}{1-(1-p)e^t}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$

If Z is a standard normal, then $X = \sigma Z + \mu$ is normal with parameters μ and σ^2; therefore,

$$\phi(t) = E[e^{tX}] = E[e^{t(\sigma Z+\mu)}] = e^{t\mu} E[e^{t\sigma Z}] = \exp\left\{\frac{\sigma^2 t^2}{2} + \mu t\right\}$$

By differentiating we obtain

$$\phi'(t) = (\mu + t\sigma^2) \exp\left\{\frac{\sigma^2 t^2}{2} + \mu t\right\},$$
$$\phi''(t) = (\mu + t\sigma^2)^2 \exp\left\{\frac{\sigma^2 t^2}{2} + \mu t\right\} + \sigma^2 \exp\left\{\frac{\sigma^2 t^2}{2} + \mu t\right\}$$

and so

$$E[X] = \phi'(0) = \mu,$$
$$E[X^2] = \phi''(0) = \mu^2 + \sigma^2$$

implying that

$$\begin{aligned}\mathrm{Var}(X) &= E[X^2] - E([X])^2 \\ &= \sigma^2\end{aligned}$$

■

Tables 2.1 and 2.2 give the moment generating function for some common distributions.

An important property of moment generating functions is that the *moment generating function of the sum of independent random variables is just the product of the individual moment generating functions*. To see this, suppose that X and Y are independent and have moment generating functions $\phi_X(t)$ and $\phi_Y(t)$, respectively. Then

Table 2.2

Continuous probability distribution	Probability density function, $f(x)$	Moment generating function, $\phi(t)$	Mean	Variance
Uniform over (a, b)	$f(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise} \end{cases}$	$\frac{e^{tb} - e^{ta}}{t(b-a)}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential with parameter $\lambda > 0$	$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$	$\frac{\lambda}{\lambda - t}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Gamma with parameters (n, λ), $\lambda > 0$	$f(x) = \begin{cases} \frac{\lambda e^{-\lambda x}(\lambda x)^{n-1}}{(n-1)!}, & x \geq 0 \\ 0, & x < 0 \end{cases}$	$\left(\frac{\lambda}{\lambda - t}\right)^n$	$\frac{n}{\lambda}$	$\frac{n}{\lambda^2}$
Normal with parameters (μ, σ^2)	$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \times \exp\{-(x-\mu)^2/2\sigma^2\}, \quad -\infty < x < \infty$	$\exp\left\{\mu t + \frac{\sigma^2 t^2}{2}\right\}$	μ	σ^2

$\phi_{X+Y}(t)$, the moment generating function of $X + Y$, is given by

$$\begin{aligned} \phi_{X+Y}(t) &= E[e^{t(X+Y)}] \\ &= E[e^{tX}e^{tY}] \\ &= E[e^{tX}]E[e^{tY}] \\ &= \phi_X(t)\phi_Y(t) \end{aligned}$$

where the next to the last equality follows from Proposition 2.3 since X and Y are independent.

Another important result is that the *moment generating function uniquely determines the distribution*. That is, there exists a one-to-one correspondence between the moment generating function and the distribution function of a random variable.

Example 2.44 (Sums of Independent Binomial Random Variables)**.** If X and Y are independent binomial random variables with parameters (n, p) and (m, p), respectively, then what is the distribution of $X + Y$?

Solution: The moment generating function of $X + Y$ is given by

$$\begin{aligned} \phi_{X+Y}(t) = \phi_X(t)\phi_Y(t) &= (pe^t + 1 - p)^n (pe^t + 1 - p)^m \\ &= (pe^t + 1 - p)^{m+n} \end{aligned}$$

But $(pe^t + (1-p))^{m+n}$ is just the moment generating function of a binomial random variable having parameters $m+n$ and p. Thus, this must be the distribution of $X+Y$. ■

Example 2.45 (Sums of Independent Poisson Random Variables)**.** Calculate the distribution of $X+Y$ when X and Y are independent Poisson random variables with means λ_1 and λ_2, respectively.

Solution:

$$\begin{aligned}\phi_{X+Y}(t) &= \phi_X(t)\phi_Y(t)\\ &= e^{\lambda_1(e^t-1)}e^{\lambda_2(e^t-1)}\\ &= e^{(\lambda_1+\lambda_2)(e^t-1)}\end{aligned}$$

Hence, $X+Y$ is Poisson distributed with mean $\lambda_1+\lambda_2$, verifying the result given in Example 2.37. ■

Example 2.46 (Sums of Independent Normal Random Variables)**.** Show that if X and Y are independent normal random variables with parameters (μ_1, σ_1^2) and (μ_2, σ_2^2), respectively, then $X+Y$ is normal with mean $\mu_1+\mu_2$ and variance $\sigma_1^2+\sigma_2^2$.

Solution:

$$\begin{aligned}\phi_{X+Y}(t) &= \phi_X(t)\phi_Y(t)\\ &= \exp\left\{\frac{\sigma_1^2t^2}{2}+\mu_1t\right\}\exp\left\{\frac{\sigma_2^2t^2}{2}+\mu_2t\right\}\\ &= \exp\left\{\frac{(\sigma_1^2+\sigma_2^2)t^2}{2}+(\mu_1+\mu_2)t\right\}\end{aligned}$$

which is the moment generating function of a normal random variable with mean $\mu_1+\mu_2$ and variance $\sigma_1^2+\sigma_2^2$. Hence, the result follows since the moment generating function uniquely determines the distribution. ■

Example 2.47 (The Poisson Paradigm)**.** We showed in Section 2.2.4 that the number of successes that occur in n independent trials, each of which results in a success with probability p is, when n is large and p small, approximately a Poisson random variable with parameter $\lambda = np$. This result, however, can be substantially strengthened. First it is not necessary that the trials have the same success probability, only that all the success probabilities are small. To see that this is the case, suppose that the trials are independent, with trial i resulting in a success with probability p_i, where all the $p_i, i = 1, \ldots, n$ are small. Letting X_i equal 1 if trial i is a success, and 0 otherwise, it follows that the number of successes, call it X, can be expressed as

$$X = \sum_{i=1}^{n} X_i$$

Using that X_i is a Bernoulli (or binary) random variable, its moment generating function is

$$E[e^{tX_i}] = p_i e^t + 1 - p_i = 1 + p_i(e^t - 1)$$

Now, using the result that, for $|x|$ small,

$$e^x \approx 1 + x$$

it follows, because $p_i(e^t - 1)$ is small when p_i is small, that

$$E[e^{tX_i}] = 1 + p_i(e^t - 1) \approx \exp\{p_i(e^t - 1)\}$$

Because the moment generating function of a sum of independent random variables is the product of their moment generating functions, the preceding implies that

$$E[e^{tX}] \approx \prod_{i=1}^{n} \exp\{p_i(e^t - 1)\} = \exp\left\{\sum_i p_i(e^t - 1)\right\}$$

But the right side of the preceding is the moment generating function of a Poisson random variable with mean $\sum_i p_i$, thus arguing that this is approximately the distribution of X.

Not only is it not necessary for the trials to have the same success probability for the number of successes to approximately have a Poisson distribution, they need not even be independent, provided that their dependence is *weak*. For instance, recall the matching problem (Example 2.30) where n people randomly select hats from a set consisting of one hat from each person. By regarding the random selections of hats as constituting n trials, where we say that trial i is a success if person i chooses his or her own hat, it follows that, with A_i being the event that trial i is a success,

$$P(A_i) = \frac{1}{n} \quad \text{and} \quad P(A_i|A_j) = \frac{1}{n-1}, \qquad j \neq i$$

Hence, whereas the trials are not independent, their dependence appears, for large n, to be weak. Because of this weak dependence, and the small trial success probabilities, it would seem that the number of matches should approximately have a Poisson distribution with mean 1 when n is large, and this is shown to be the case in Example 3.27.

The statement that "the number of successes in n trials that are either independent or at most weakly dependent is, when the trial success probabilities are all small, approximately a Poisson random variable" is known as the *Poisson paradigm*. ■

Remark. For a nonnegative random variable X, it is often convenient to define its *Laplace transform* $g(t), t \geq 0$, by

$$g(t) = \phi(-t) = E[e^{-tX}]$$

That is, the Laplace transform evaluated at t is just the moment generating function evaluated at $-t$. The advantage of dealing with the Laplace transform, rather than the

moment generating function, when the random variable is nonnegative is that if $X \geq 0$ and $t \geq 0$, then

$$0 \leq e^{-tX} \leq 1$$

That is, the Laplace transform is always between 0 and 1. As in the case of moment generating functions, it remains true that nonnegative random variables that have the same Laplace transform must also have the same distribution. ■

It is also possible to define the joint moment generating function of two or more random variables. This is done as follows. For any n random variables $X_1, \ldots, X_n$, the joint moment generating function, $\phi(t_1, \ldots, t_n)$, is defined for all real values of $t_1, \ldots, t_n$ by

$$\phi(t_1, \ldots, t_n) = E[e^{(t_1 X_1 + \cdots + t_n X_n)}]$$

It can be shown that $\phi(t_1, \ldots, t_n)$ uniquely determines the joint distribution of $X_1, \ldots, X_n$.

Example 2.48 (The Multivariate Normal Distribution)**.** Let $Z_1, \ldots, Z_n$ be a set of n independent standard normal random variables. If, for some constants a_{ij}, $1 \leq i \leq m$, $1 \leq j \leq n$, and μ_i, $1 \leq i \leq m$,

$$\begin{aligned}
X_1 &= a_{11} Z_1 + \cdots + a_{1n} Z_n + \mu_1, \\
X_2 &= a_{21} Z_1 + \cdots + a_{2n} Z_n + \mu_2, \\
&\vdots \\
X_i &= a_{i1} Z_1 + \cdots + a_{in} Z_n + \mu_i, \\
&\vdots \\
X_m &= a_{m1} Z_1 + \cdots + a_{mn} Z_n + \mu_m
\end{aligned}$$

then the random variables $X_1, \ldots, X_m$ are said to have a multivariate normal distribution.

It follows from the fact that the sum of independent normal random variables is itself a normal random variable that each X_i is a normal random variable with mean and variance given by

$$\begin{aligned}
E[X_i] &= \mu_i, \\
\mathrm{Var}(X_i) &= \sum_{j=1}^{n} a_{ij}^2
\end{aligned}$$

Let us now determine

$$\phi(t_1, \ldots, t_m) = E[\exp\{t_1 X_1 + \cdots + t_m X_m\}]$$

the joint moment generating function of $X_1, \ldots, X_m$. The first thing to note is that since $\sum_{i=1}^m t_i X_i$ is itself a linear combination of the independent normal random variables $Z_1, \ldots, Z_n$, it is also normally distributed. Its mean and variance are respectively

$$E\left[\sum_{i=1}^m t_i X_i\right] = \sum_{i=1}^m t_i \mu_i$$

and

$$\begin{aligned}\text{Var}\left(\sum_{i=1}^m t_i X_i\right) &= \text{Cov}\left(\sum_{i=1}^m t_i X_i, \sum_{j=1}^m t_j X_j\right) \\ &= \sum_{i=1}^m \sum_{j=1}^m t_i t_j \text{Cov}(X_i, X_j)\end{aligned}$$

Now, if Y is a normal random variable with mean μ and variance σ^2, then

$$E[e^Y] = \phi_Y(t)|_{t=1} = e^{\mu+\sigma^2/2}$$

Thus, we see that

$$\phi(t_1, \ldots, t_m) = \exp\left\{\sum_{i=1}^m t_i \mu_i + \frac{1}{2}\sum_{i=1}^m \sum_{j=1}^m t_i t_j \text{Cov}(X_i, X_j)\right\}$$

which shows that the joint distribution of $X_1, \ldots, X_m$ is completely determined from a knowledge of the values of $E[X_i]$ and $\text{Cov}(X_i, X_j)$, $i, j = 1, \ldots, m$. ■

2.6.1 The Joint Distribution of the Sample Mean and Sample Variance from a Normal Population

Let $X_1, \ldots, X_n$ be independent and identically distributed random variables, each with mean μ and variance σ^2. The random variable S^2 defined by

$$S^2 = \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{n-1}$$

is called the *sample variance* of these data. To compute $E[S^2]$ we use the identity

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2 \tag{2.21}$$

which is proven as follows:

$$\sum_{i=1}^n (X_i - \bar{X}) = \sum_{i=1}^n (X_i - \mu + \mu - \bar{X})^2$$

$$
\begin{aligned}
&= \sum_{i=1}^{n} (X_i - \mu)^2 + n(\mu - \bar{X})^2 + 2(\mu - \bar{X}) \sum_{i=1}^{n} (X_i - \mu) \\
&= \sum_{i=1}^{n} (X_i - \mu)^2 + n(\mu - \bar{X})^2 + 2(\mu - \bar{X})(n\bar{X} - n\mu) \\
&= \sum_{i=1}^{n} (X_i - \mu)^2 + n(\mu - \bar{X})^2 - 2n(\mu - \bar{X})^2
\end{aligned}
$$

and Identity (2.21) follows.

Using Identity (2.21) gives

$$
\begin{aligned}
E[(n-1)S^2] &= \sum_{i=1}^{n} E[(X_i - \mu)^2] - nE[(\bar{X} - \mu)^2] \\
&= n\sigma^2 - n \operatorname{Var}(\bar{X}) \\
&= (n-1)\sigma^2 \qquad \text{from Proposition 2.4(b)}
\end{aligned}
$$

Thus, we obtain from the preceding that

$$E[S^2] = \sigma^2$$

We will now determine the joint distribution of the sample mean $\bar{X} = \sum_{i=1}^{n} X_i/n$ and the sample variance S^2 when the X_i have a normal distribution. To begin we need the concept of a chi-squared random variable.

Definition 2.2. If $Z_1, \ldots, Z_n$ are independent standard normal random variables, then the random variable $\sum_{i=1}^{n} Z_i^2$ is said to be a *chi-squared random variable* with n *degrees of freedom*.

We shall now compute the moment generating function of $\sum_{i=1}^{n} Z_i^2$. To begin, note that

$$
\begin{aligned}
E[\exp\{tZ_i^2\}] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx^2} e^{-x^2/2}\, dx \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2/2\sigma^2}\, dx \qquad \text{where } \sigma^2 = (1-2t)^{-1} \\
&= \sigma \\
&= (1-2t)^{-1/2}
\end{aligned}
$$

Hence,

$$E\left[\exp\left\{t \sum_{i=1}^{n} Z_i^2\right\}\right] = \prod_{i=1}^{n} E[\exp\{tZ_i^2\}] = (1-2t)^{-n/2}$$

Now, let $X_1, \ldots, X_n$ be independent normal random variables, each with mean μ and variance σ^2, and let $\bar{X} = \sum_{i=1}^{n} X_i/n$ and S^2 denote their sample mean and sample

variance. Since the sum of independent normal random variables is also a normal random variable, it follows that $\bar{X}$ is a normal random variable with expected value μ and variance σ^2/n. In addition, from Proposition 2.4,

$$\text{Cov}(\bar{X}, X_i - \bar{X}) = 0, \qquad i = 1, \ldots, n \tag{2.22}$$

Also, since $\bar{X}, X_1 - \bar{X}, X_2 - \bar{X}, \ldots, X_n - \bar{X}$ are all linear combinations of the independent standard normal random variables $(X_i - \mu)/\sigma, i = 1, \ldots, n$, it follows that the random variables $\bar{X}, X_1 - \bar{X}, X_2 - \bar{X}, \ldots, X_n - \bar{X}$ have a joint distribution that is multivariate normal. However, if we let Y be a normal random variable with mean μ and variance σ^2/n that is independent of $X_1, \ldots, X_n$, then the random variables $Y, X_1 - \bar{X}, X_2 - \bar{X}, \ldots, X_n - \bar{X}$ also have a multivariate normal distribution, and by Eq. (2.22), they have the same expected values and covariances as the random variables $\bar{X}, X_i - \bar{X}, i = 1, \ldots, n$. Thus, since a multivariate normal distribution is completely determined by its expected values and covariances, we can conclude that the random vectors $Y, X_1 - \bar{X}, X_2 - \bar{X}, \ldots, X_n - \bar{X}$ and $\bar{X}, X_1 - \bar{X}, X_2 - \bar{X}, \ldots, X_n - \bar{X}$ have the same joint distribution; thus showing that $\bar{X}$ is independent of the sequence of deviations $X_i - \bar{X}, i = 1, \ldots, n$.

Since $\bar{X}$ is independent of the sequence of deviations $X_i - \bar{X}, i = 1, \ldots, n$, it follows that it is also independent of the sample variance

$$S^2 \equiv \sum_{i=1}^{n} \frac{(X_i - \bar{X})^2}{n-1}$$

To determine the distribution of S^2, use Identity (2.21) to obtain

$$(n-1)S^2 = \sum_{i=1}^{n} (X_i - \mu)^2 - n(\bar{X} - \mu)^2$$

Dividing both sides of this equation by σ^2 yields

$$\frac{(n-1)S^2}{\sigma^2} + \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}\right)^2 = \sum_{i=1}^{n} \frac{(X_i - \mu)^2}{\sigma^2} \tag{2.23}$$

Now, $\sum_{i=1}^{n} (X_i - \mu)^2/\sigma^2$ is the sum of the squares of n independent standard normal random variables, and so is a chi-squared random variable with n degrees of freedom; it thus has moment generating function $(1 - 2t)^{-n/2}$. Also $[(\bar{X} - \mu)/(\sigma/\sqrt{n})]^2$ is the square of a standard normal random variable and so is a chi-squared random variable with one degree of freedom; it thus has moment generating function $(1 - 2t)^{-1/2}$. In addition, we have previously seen that the two random variables on the left side of Eq. (2.23) are independent. Therefore, because the moment generating function of the sum of independent random variables is equal to the product of their individual moment generating functions, we obtain that

$$E[e^{t(n-1)S^2/\sigma^2}](1 - 2t)^{-1/2} = (1 - 2t)^{-n/2}$$

or

$$E[e^{t(n-1)S^2/\sigma^2}] = (1-2t)^{-(n-1)/2}$$

But because $(1-2t)^{-(n-1)/2}$ is the moment generating function of a chi-squared random variable with $n-1$ degrees of freedom, we can conclude, since the moment generating function uniquely determines the distribution of the random variable, that this is the distribution of $(n-1)S^2/\sigma^2$.

Summing up, we have shown the following.

Proposition 2.5. *If $X_1, \ldots, X_n$ are independent and identically distributed normal random variables with mean μ and variance σ^2, then the sample mean $\bar{X}$ and the sample variance S^2 are independent. $\bar{X}$ is a normal random variable with mean μ and variance σ^2/n; $(n-1)S^2/\sigma^2$ is a chi-squared random variable with $n-1$ degrees of freedom.*

2.7 Limit Theorems

We start this section by proving a result known as Markov's inequality.

Proposition 2.6 (Markov's Inequality)**.** *If X is a random variable that takes only nonnegative values, then for any value $a > 0$*

$$P\{X \geq a\} \leq \frac{E[X]}{a}$$

Proof. We give a proof for the case where X is continuous with density f.

$$\begin{aligned}
E[X] &= \int_0^\infty x f(x)\,dx \\
&= \int_0^a x f(x)\,dx + \int_a^\infty x f(x)\,dx \\
&\geq \int_a^\infty x f(x)\,dx \\
&\geq \int_a^\infty a f(x)\,dx \\
&= a \int_a^\infty f(x)\,dx \\
&= a P\{X \geq a\}
\end{aligned}$$

and the result is proven. ■

As a corollary, we obtain the following.

Proposition 2.7 (Chebyshev's Inequality)**.** *If X is a random variable with mean μ and variance σ^2, then, for any value $k > 0$,*

$$P\{|X - \mu| \geq k\} \leq \frac{\sigma^2}{k^2}$$

Proof. Since $(X - \mu)^2$ is a nonnegative random variable, we can apply Markov's inequality (with $a = k^2$) to obtain

$$P\{(X - \mu)^2 \geq k^2\} \leq \frac{E[(X - \mu)^2]}{k^2}$$

But since $(X - \mu)^2 \geq k^2$ if and only if $|X - \mu| \geq k$, the preceding is equivalent to

$$P\{|X - \mu| \geq k\} \leq \frac{E[(X - \mu)^2]}{k^2} = \frac{\sigma^2}{k^2}$$

and the proof is complete. ■

The importance of Markov's and Chebyshev's inequalities is that they enable us to derive bounds on probabilities when only the mean, or both the mean and the variance, of the probability distribution are known. Of course, if the actual distribution were known, then the desired probabilities could be exactly computed, and we would not need to resort to bounds.

Example 2.49. Suppose we know that the number of items produced in a factory during a week is a random variable with mean 500.

(a) What can be said about the probability that this week's production will be at least 1000?

(b) If the variance of a week's production is known to equal 100, then what can be said about the probability that this week's production will be between 400 and 600?

Solution: Let X be the number of items that will be produced in a week.

(a) By Markov's inequality,

$$P\{X \geq 1000\} \leq \frac{E[X]}{1000} = \frac{500}{1000} = \frac{1}{2}$$

(b) By Chebyshev's inequality,

$$P\{|X - 500| \geq 100\} \leq \frac{\sigma^2}{(100)^2} = \frac{1}{100}$$

Hence,

$$P\{|X - 500| < 100\} \geq 1 - \frac{1}{100} = \frac{99}{100}$$

and so the probability that this week's production will be between 400 and 600 is at least 0.99. ■

The following theorem, known as the *strong law of large numbers*, is probably the most well-known result in probability theory. It states that the average of a sequence of independent random variables having the same distribution will, with probability 1, converge to the mean of that distribution.

Theorem 2.1 (Strong Law of Large Numbers). *Let $X_1, X_2, \ldots$ be a sequence of independent random variables having a common distribution, and let $E[X_i] = \mu$. Then, with probability 1,*

$$\frac{X_1 + X_2 + \cdots + X_n}{n} \to \mu \qquad \text{as } n \to \infty$$

As an example of the preceding, suppose that a sequence of independent trials is performed. Let E be a fixed event and denote by $P(E)$ the probability that E occurs on any particular trial. Letting

$$X_i = \begin{cases} 1, & \text{if } E \text{ occurs on the } i\text{th trial} \\ 0, & \text{if } E \text{ does not occur on the } i\text{th trial} \end{cases}$$

we have by the strong law of large numbers that, with probability 1,

$$\frac{X_1 + \cdots + X_n}{n} \to E[X] = P(E) \tag{2.24}$$

Since $X_1 + \cdots + X_n$ represents the number of times that the event E occurs in the first n trials, we may interpret Eq. (2.24) as stating that, with probability 1, the limiting proportion of time that the event E occurs is just $P(E)$.

Running neck and neck with the strong law of large numbers for the honor of being probability theory's number one result is the *central limit theorem*. Besides its theoretical interest and importance, this theorem provides a simple method for computing approximate probabilities for sums of independent random variables. It also explains the remarkable fact that the empirical frequencies of so many natural "populations" exhibit a bell-shaped (that is, normal) curve.

Theorem 2.2 (Central Limit Theorem). *Let $X_1, X_2, \ldots$ be a sequence of independent, identically distributed random variables, each with mean μ and variance σ^2. Then the distribution of*

$$\frac{X_1 + X_2 + \cdots + X_n - n\mu}{\sigma\sqrt{n}}$$

tends to the standard normal as $n \to \infty$. That is,

$$P\left\{\frac{X_1 + X_2 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} \le a\right\} \to \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{a} e^{-x^2/2}\, dx$$

as $n \to \infty$.

Note that like the other results of this section, this theorem holds for *any* distribution of the X_is; herein lies its power.

If X is binomially distributed with parameters n and p, then X has the same distribution as the sum of n independent Bernoulli random variables, each with parameter p. (Recall that the Bernoulli random variable is just a binomial random variable whose parameter n equals 1.) Hence, the distribution of

$$\frac{X - E[X]}{\sqrt{\text{Var}(X)}} = \frac{X - np}{\sqrt{np(1-p)}}$$

approaches the standard normal distribution as n approaches ∞. The normal approximation will, in general, be quite good for values of n satisfying $np(1-p) \geq 10$.

Example 2.50 (Normal Approximation to the Binomial)**.** Let X be the number of times that a fair coin, flipped 40 times, lands heads. Find the probability that $X = 20$. Use the normal approximation and then compare it to the exact solution.

Solution: Since the binomial is a discrete random variable, and the normal a continuous random variable, it leads to a better approximation to write the desired probability as

$$\begin{aligned} P\{X = 20\} &= P\{19.5 < X < 20.5\} \\ &= P\left\{\frac{19.5 - 20}{\sqrt{10}} < \frac{X - 20}{\sqrt{10}} < \frac{20.5 - 20}{\sqrt{10}}\right\} \\ &= P\left\{-0.16 < \frac{X - 20}{\sqrt{10}} < 0.16\right\} \\ &\approx \Phi(0.16) - \Phi(-0.16) \end{aligned}$$

where $\Phi(x)$, the probability that the standard normal is less than x is given by

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-y^2/2}\, dy$$

By the symmetry of the standard normal distribution

$$\Phi(-0.16) = P\{N(0,1) > 0.16\} = 1 - \Phi(0.16)$$

where $N(0,1)$ is a standard normal random variable. Hence, the desired probability is approximated by

$$P\{X = 20\} \approx 2\Phi(0.16) - 1$$

Using Table 2.3, we obtain

$$P\{X = 20\} \approx 0.1272$$

Table 2.3 Area $\Phi(x)$ under the Standard Normal Curve to the Left of x.

x	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5597	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8557	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

The exact result is

$$P\{X = 20\} = \binom{40}{20}\left(\frac{1}{2}\right)^{40}$$

which can be shown to equal 0.1254. ■

Example 2.51. Let $X_i, i = 1, 2, \ldots, 10$ be independent random variables, each being uniformly distributed over (0, 1). Estimate $P\{\sum_1^{10} X_i > 7\}$.

Solution: Since $E[X_i] = \frac{1}{2}$, $\text{Var}(X_i) = \frac{1}{12}$ we have by the central limit theorem that

$$\begin{aligned} P\left\{\sum_1^{10} X_i > 7\right\} &= P\left\{\frac{\sum_1^{10} X_i - 5}{\sqrt{10\left(\frac{1}{12}\right)}} > \frac{7-5}{\sqrt{10\left(\frac{1}{12}\right)}}\right\} \\ &\approx 1 - \Phi(2.19) \\ &= 0.0143 \end{aligned}$$

■

Example 2.52. The lifetime of a special type of battery is a random variable with mean 40 hours and standard deviation 20 hours. A battery is used until it fails, at which point it is replaced by a new one. Assuming a stockpile of 25 such batteries, the lifetimes of which are independent, approximate the probability that over 1100 hours of use can be obtained.

Solution: If we let X_i denote the lifetime of the ith battery to be put in use, then we desire $p = P\{X_1 + \cdots + X_{25} > 1100\}$, which is approximated as follows:

$$\begin{aligned} p &= P\left\{\frac{X_1 + \cdots + X_{25} - 1000}{20\sqrt{25}} > \frac{1100 - 1000}{20\sqrt{25}}\right\} \\ &\approx P\{N(0, 1) > 1\} \\ &= 1 - \Phi(1) \\ &\approx 0.1587 \end{aligned}$$

■

We now present a heuristic proof of the central limit theorem. Suppose first that the X_i have mean 0 and variance 1, and let $E[e^{tX}]$ denote their common moment generating function. Then, the moment generating function of $\frac{X_1+\cdots+X_n}{\sqrt{n}}$ is

$$\begin{aligned} E\left[\exp\left\{t\left(\frac{X_1 + \cdots + X_n}{\sqrt{n}}\right)\right\}\right] &= E[e^{tX_1/\sqrt{n}} e^{tX_2/\sqrt{n}} \cdots e^{tX_n/\sqrt{n}}] \\ &= (E[e^{tX/\sqrt{n}}])^n \qquad \text{by independence} \end{aligned}$$

Now, for n large, we obtain from the Taylor series expansion of e^y that

$$e^{tX/\sqrt{n}} \approx 1 + \frac{tX}{\sqrt{n}} + \frac{t^2X^2}{2n}$$

Taking expectations shows that when n is large

$$E[e^{tX/\sqrt{n}}] \approx 1 + \frac{tE[X]}{\sqrt{n}} + \frac{t^2E[X^2]}{2n}$$
$$= 1 + \frac{t^2}{2n} \qquad \text{because } E[X] = 0, E[X^2] = 1$$

Therefore, we obtain that when n is large

$$E\left[\exp\left\{t\left(\frac{X_1 + \cdots + X_n}{\sqrt{n}}\right)\right\}\right] \approx \left(1 + \frac{t^2}{2n}\right)^n$$

When n goes to ∞ the approximation can be shown to become exact and we have

$$\lim_{n\to\infty} E\left[\exp\left\{t\left(\frac{X_1 + \cdots + X_n}{\sqrt{n}}\right)\right\}\right] = e^{t^2/2}$$

Thus, the moment generating function of $\frac{X_1+\cdots+X_n}{\sqrt{n}}$ converges to the moment generating function of a (standard) normal random variable with mean 0 and variance 1. Using this, it can be proven that the distribution function of the random variable $\frac{X_1+\cdots+X_n}{\sqrt{n}}$ converges to the standard normal distribution function Φ.

When the X_i have mean μ and variance σ^2, the random variables $\frac{X_i-\mu}{\sigma}$ have mean 0 and variance 1. Thus, the preceding shows that

$$P\left\{\frac{X_1 - \mu + X_2 - \mu + \cdots + X_n - \mu}{\sigma\sqrt{n}} \le a\right\} \to \Phi(a)$$

which proves the central limit theorem.

2.8 Proof of the Strong Law of Large Numbers

In this section we give a proof of the strong law of large numbers. Our proof of the strong law makes use of the *Borel–Cantelli lemma*.

Borel–Cantelli Lemma. *For a sequence of events A_i, $i \ge 1$, let N denote the number of these events that occur. If $\sum_{i=1}^{\infty} P(A_i) < \infty$, then $P(N = \infty) = 0$.*

Proof. Suppose that $\sum_{i=1}^{\infty} P(A_i) < \infty$. Now, if $N = \infty$, then for every $n < \infty$ at least one of the events $A_n, A_{n+1}, \ldots$ will occur. That is, $N = \infty$ implies that $\cup_{i=n}^{\infty} A_i$ occurs for every n. Thus, for every n

$$P(N = \infty) \le P(\cup_{i=n}^{\infty} A_i)$$
$$\le \sum_{i=n}^{\infty} P(A_i)$$

where the final inequality follows from Boole's inequality. Because $\sum_{i=1}^{\infty} P(A_i) < \infty$ implies that $\sum_{i=n}^{\infty} P(A_i) \to 0$ as $n \to \infty$, we obtain from the preceding upon letting $n \to \infty$ that $P(N = \infty) = 0$, which proves the result. ■

Remark. The Borel–Cantelli lemma is actually quite intuitive, for if we define the indicator variable I_i to equal 1 if A_i occurs and to equal 0 otherwise, then $N = \sum_{i=1}^{\infty} I_i$, implying that

$$E[N] = \sum_{i=1}^{\infty} E[I_i] = \sum_{i=1}^{\infty} P(A_i)$$

Consequently, the Borel–Cantelli theorem states that if the expected number of events that occur is finite then the probability that an infinite number of them occur is 0, which is intuitive because if there were a positive probability that an infinite number of events could occur then $E[N]$ would be infinite. ■

Suppose that $X_1, X_2, \ldots$ are independent and identically distributed random variables with mean μ, and let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$ be the average of the first n of them. The strong law of large numbers states that $P(\lim_{n\to\infty} \bar{X}_n = \mu) = 1$. That is, with probability 1, $\bar{X}_n$ converges to μ as $n \to \infty$. We will give a proof of this result under the assumption that σ^2, the variance of X_i, is finite (which is equivalent to assuming that $E[X_i^2] < \infty$). Because proving the strong law requires showing, for any $\epsilon > 0$, that $|\bar{X}_n - \mu| > \epsilon$ for only a finite number of values of n, it is natural to attempt to prove it by utilizing the Borel–Cantelli lemma. That is, the result would follow if we could show that $\sum_{n=1}^{\infty} P(|\bar{X}_n - \mu| > \epsilon) < \infty$. However, because $E[\bar{X}_n] = \mu$, $\text{Var}(\bar{X}_n) = \sigma^2/n$, attempting to show this by using Chebyshev's inequality yields

$$\sum_{n=1}^{\infty} P(|\bar{X}_n - \mu| > \epsilon) \le \sum_{n=1}^{\infty} \frac{\text{Var}(\bar{X}_n)}{\epsilon^2} = \frac{\sigma^2}{\epsilon^2} \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

Thus, a straightforward use of Borel–Cantelli does not work. However, as we now show, a tweaking of the argument, where we first consider a subsequence of $\bar{X}_n, n \ge 1$, allows us to prove the strong law.

Theorem (The Strong Law of Large Numbers)**.** *Let $X_1, X_2, \ldots$ be a sequence of independent and identically distributed random variables with $E[X_i] = \mu$, and $Var(X_i) = \sigma^2 < \infty$. Then, with $\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i$,*

$$P(\lim_{n\to\infty} \bar{X}_n = \mu) = 1$$

Proof. Suppose first that the X_i are nonnegative random variables. Fix $\alpha > 1$, and let n_j be the smallest integer greater than or equal to α^j, $j \ge 1$. From Chebyshev's inequality we see that

$$P(|\bar{X}_{n_j} - \mu| > \epsilon) \le \frac{\text{Var}(\bar{X}_{n_j})}{\epsilon^2} = \frac{\sigma^2}{n_j \epsilon^2}$$

Consequently,

$$\sum_{j=1}^{\infty} P(|\bar{X}_{n_j} - \mu| > \epsilon) \leq \frac{\sigma^2}{\epsilon^2} \sum_{j=1}^{\infty} \frac{1}{n_j} \leq \frac{\sigma^2}{\epsilon^2} \sum_{j=1}^{\infty} (1/\alpha)^j < \infty.$$

Therefore, by the Borel–Cantelli lemma, it follows that, with probability 1, $|\bar{X}_{n_j} - \mu| > \epsilon$ for only a finite number of j. As this is true for any $\epsilon > 0$, we see that, with probability 1,

$$\lim_{j \to \infty} \bar{X}_{n_j} = \mu \tag{2.25}$$

Because $n_j \uparrow \infty$ as $j \uparrow \infty$, it follows that for any $m > \alpha$, there is an integer $j(m)$ such that $n_{j(m)} \leq m < n_{j(m)+1}$. The nonnegativity of the X_i yields that

$$\sum_{i=1}^{n_{j(m)}} X_i \leq \sum_{i=1}^{m} X_i \leq \sum_{i=1}^{n_{j(m)+1}} X_i$$

Dividing by m shows that

$$\frac{n_{j(m)}}{m} \bar{X}_{n_{j(m)}} \leq \bar{X}_m \leq \frac{n_{j(m)+1}}{m} \bar{X}_{n_{j(m)+1}}$$

Because $\frac{1}{n_{j(m)+1}} < \frac{1}{m} \leq \frac{1}{n_{j(m)}}$, this yields that

$$\frac{n_{j(m)}}{n_{j(m)+1}} \bar{X}_{n_{j(m)}} \leq \bar{X}_m \leq \frac{n_{j(m)+1}}{n_{j(m)}} \bar{X}_{n_{j(m)+1}}$$

Because $\lim_{m \to \infty} j(m) = \infty$ and $\lim_{j \to \infty} \frac{n_{j+1}}{n_j} = \alpha$, it follows, for any $\epsilon > 0$, that $\frac{n_{j(m)+1}}{n_{j(m)}} < \alpha + \epsilon$ for all but a finite number of m. Consequently, from (2.25) and the preceding, it follows, with probability 1, that $\frac{\mu}{\alpha+\epsilon} < \bar{X}_m < (\alpha + \epsilon)\mu$ for all but a finite number of values of m. As this is true for any $\epsilon > 0$, $\alpha > 1$, it follows that with probability 1

$$\lim_{m \to \infty} \bar{X}_m = \mu$$

Thus the result is proven when the X_i are nonnegative. In the general case, let

$$X_i^+ = \begin{cases} X_i, & \text{if } X_i \geq 0 \\ 0, & \text{if } X_i < 0 \end{cases}$$

and let

$$X_i^- = \begin{cases} 0, & \text{if } X_i \geq 0 \\ -X_i, & \text{if } X_i < 0 \end{cases}$$

X_i^+ and X_i^- are called, respectively, the positive and negative parts of X_i. Noting that

$$X_i = X_i^+ - X_i^-$$

it follows (since $X_i^+ X_i^- = 0$) that

$$X_i^2 = (X_i^+)^2 + (X_i^-)^2$$

Hence, the assumption that $E[X_i^2] < \infty$ implies that $E[(X_i^+)^2]$ and $E[(X_i^-)^2]$ are also both finite. Letting $\mu^+ = E[X_i^+]$ and $\mu^- = E[X_i^-]$, and using that X_i^+ and X_i^- are both nonnegative, it follows from the previous result for nonnegative random variables that, with probability 1,

$$\lim_{m\to\infty} \frac{1}{m}\sum_{i=1}^{m} X_i^+ = \mu^+, \quad \lim_{m\to\infty} \frac{1}{m}\sum_{i=1}^{m} X_i^- = \mu^-$$

Consequently, with probability 1,

$$\begin{aligned}\lim_{m\to\infty} \bar{X}_m &= \lim_{m\to\infty} \frac{1}{m}\sum_{i=1}^{m}(X_i^+ - X_i^-)\\ &= \mu^+ - \mu^-\\ &= \mu\end{aligned}$$

■

There is a partial converse of the Borel–Cantelli lemma that holds when the events are independent.

Converse to Borel–Cantelli Lemma. *If $\sum_{i=1}^{\infty} P(A_i) = \infty$, and the events A_i, $i \geq 1$ are independent, then*

$$P(\textit{an infinite number of the events } A_i,\ i \geq 1 \textit{ occur}) = 1$$

Proof. For any n, let $B_n = \cap_{i=n}^{\infty} A_i^c$ be the event that none the events $A_n, A_{n+1}, \ldots$ occur. Then

$$\begin{aligned}P(B_n) &= P(\cap_{i=n}^{\infty} A_i^c)\\ &= \prod_{i=n}^{\infty} P(A_i^c) \qquad \text{by independence}\\ &= \prod_{i=n}^{\infty}[1 - P(A_i)]\\ &\leq \prod_{i=n}^{\infty} e^{-P(A_i)} \qquad \text{by the inequality } e^{-x} \geq 1 - x\\ &= e^{-\sum_{i=n}^{\infty} P(A_i)}\\ &= 0\end{aligned}$$

Because B_n, $n \geq 1$ are increasing events, $\lim_{n\to\infty} B_n = \cup_{n=1}^{\infty} B_n$. Consequently, it follows from the continuity property of probabilities that

$$\begin{aligned} P(\cup_{n=1}^{\infty} B_n) &= P(\lim_{n\to\infty} B_n) \\ &= \lim_{n\to\infty} P(B_n) \\ &= 0 \end{aligned}$$

As $\cup_{n=1}^{\infty} B_n$ is the event that only a finite number of the events A_i occur, the result follows. ■

Example 2.53. Suppose that in each time period we must choose one of n drugs to use, with drug i resulting in a cure with unknown probability p_i, $i = 1, \ldots, n$. Assume that the result of a drug choice (either a cure or not) is immediately learned. Say that a drug is optimal if its cure probability is equal to $\max_i p_i$, and say that it is non-optimal otherwise. Suppose that our objective is to find a policy for deciding which drug to prescribe at each period that has the property that its use results in the long run proportion of time that a non-optimal drug is used being equal to 0. The following policy accomplishes this goal.

Suppose at time k that previous uses of drug i have resulted in $s_i(k)$ cures and $f_i(k)$ failures, for $i = 1, \ldots, n$, where $\sum_i (s_i(k) + f_i(k)) = k - 1$. Let the next choice be a "random choice" with probability $1/k$, or a "non-random choice" with probability $1 - 1/k$. If it is a random choice, let the drug used in period k be equally likely to be any of the n drugs; if it is a non-random choice, let the drug used in period k be any of the drugs with maximal value of $\frac{s_i(k)}{s_i(k)+f_i(k)}$.

To show that the use of the preceding procedure results in the long run proportion of time that a non-optimal drug is chosen being equal to 0, first note that the converse to the Borel–Cantelli lemma shows that, with probability 1, the number of random choices is infinite. As each such choice is equally likely to be any of the n drugs, it thus follows that each drug is, with probability 1, chosen infinitely often. Thus, by the strong law of large numbers, it follows that with probability 1

$$\lim_{k\to\infty} \frac{s_i(k)}{s_i(k) + f_i(k)} = p_i, \quad i = 1, \ldots, n$$

Hence, after some finite time no non-optimal drug will ever be selected by a non-random choice.

To complete the argument we now show that, with probability 1, the long run fraction of choices that are random is equal to 0. Suppose that these choices are determined by letting $U_k, k \geq 1$ be independent uniform $(0, 1)$ random variables, and then letting the choice at time k be random if $U_k \leq 1/k$. Then, with $I\{A\}$ being the indicator variable for the event A, equal to 1 if A occurs and to 0 otherwise, we have that for any m

$$\begin{aligned}\text{proportion of choices that are random} &= \lim_{r\to\infty} \frac{\sum_{k=1}^{r} I\{U_k \le 1/k\}}{r} \\ &= \lim_{r\to\infty} \frac{\sum_{k=m}^{m+r-1} I\{U_k \le 1/k\}}{r} \\ &\le \lim_{r\to\infty} \frac{\sum_{k=m}^{m+r-1} I\{U_k \le 1/m\}}{r} \\ &= 1/m\end{aligned}$$

where the next to last equality follows because if $k \ge m$ then $U_k \le 1/k \Rightarrow U_k \le 1/m$, and the final equality follows from the strong law of large numbers because $I\{U_k \le 1/m\}$, $k \ge m$, are independent and identically distributed Bernoulli random variables with mean $1/m$. As the preceding is true for all m, it follows that the proportion of choices that are random is equal to 0. It now follows from the earlier result that the long run proportion of non-random choices that are optimal is 1, that the long run proportion of choices that are optimal is equal to 1. ■

2.9 Stochastic Processes

A *stochastic process* $\{X(t), t \in T\}$ is a collection of random variables. That is, for each $t \in T$, $X(t)$ is a random variable. The index t is often interpreted as time and, as a result, we refer to $X(t)$ as the *state* of the process at time t. For example, $X(t)$ might equal the total number of customers that have entered a supermarket by time t; or the number of customers in the supermarket at time t; or the total amount of sales that have been recorded in the market by time t; etc.

The set T is called the *index* set of the process. When T is a countable set the stochastic process is said to be a *discrete-time* process. If T is an interval of the real line, the stochastic process is said to be a *continuous-time* process. For instance, $\{X_n, n = 0, 1, \ldots\}$ is a discrete-time stochastic process indexed by the nonnegative integers; while $\{X(t), t \ge 0\}$ is a continuous-time stochastic process indexed by the nonnegative real numbers.

The *state space* of a stochastic process is defined as the set of all possible values that the random variables $X(t)$ can assume.

Thus, a stochastic process is a family of random variables that describes the evolution through time of some (physical) process. We shall see much of stochastic processes in the following chapters of this text.

Example 2.54. Consider a particle that moves along a set of $m + 1$ nodes, labeled $0, 1, \ldots, m$, that are arranged around a circle (see Fig. 2.3). At each step the particle is equally likely to move one position in either the clockwise or counterclockwise direction. That is, if X_n is the position of the particle after its nth step then

$$P\{X_{n+1} = i + 1 | X_n = i\} = P\{X_{n+1} = i - 1 | X_n = i\} = \tfrac{1}{2}$$

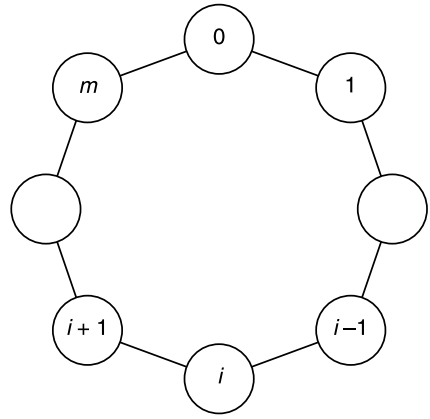


Figure 2.3 Particle moving around a circle.

where $i + 1 \equiv 0$ when $i = m$, and $i - 1 \equiv m$ when $i = 0$. Suppose now that the particle starts at 0 and continues to move around according to the preceding rules until all the nodes $1, 2, \ldots, m$ have been visited. What is the probability that node i, $i = 1, \ldots, m$, is the last one visited?

Solution: Surprisingly enough, the probability that node i is the last node visited can be determined without any computations. To do so, consider the first time that the particle is at one of the two neighbors of node i, that is, the first time that the particle is at one of the nodes $i - 1$ or $i + 1$ (with $m + 1 \equiv 0$). Suppose it is at node $i - 1$ (the argument in the alternative situation is identical). Since neither node i nor $i + 1$ has yet been visited, it follows that i will be the last node visited if and only if $i + 1$ is visited before i. This is so because in order to visit $i + 1$ before i the particle will have to visit all the nodes on the counterclockwise path from $i - 1$ to $i + 1$ before it visits i. But the probability that a particle at node $i - 1$ will visit $i + 1$ before i is just the probability that a particle will progress $m - 1$ steps in a specified direction before progressing one step in the other direction. That is, it is equal to the probability that a gambler who starts with one unit, and wins one when a fair coin turns up heads and loses one when it turns up tails, will have his fortune go up by $m - 1$ before he goes broke. Hence, because the preceding implies that the probability that node i is the last node visited is the same for all i, and because these probabilities must sum to 1, we obtain

$$P\{i \text{ is the last node visited}\} = 1/m, \qquad i = 1, \ldots, m$$ ■

Remark. The argument used in Example 2.54 also shows that a gambler who is equally likely to either win or lose one unit on each gamble will be down n before being up 1 with probability $1/(n + 1)$; or equivalently,

$$P\{\text{gambler is up 1 before being down } n\} = \frac{n}{n+1}$$

Suppose now we want the probability that the gambler is up 2 before being down n. Upon conditioning on whether he reaches up 1 before down n, we obtain that

$$\begin{aligned} &P\{\text{gambler is up 2 before being down } n\} \\ &\quad = P\{\text{up 2 before down } n | \text{up 1 before down } n\}\frac{n}{n+1} \\ &\quad = P\{\text{up 1 before down } n+1\}\frac{n}{n+1} \\ &\quad = \frac{n+1}{n+2}\frac{n}{n+1} = \frac{n}{n+2} \end{aligned}$$

Repeating this argument yields that

$$P\{\text{gambler is up } k \text{ before being down } n\} = \frac{n}{n+k}$$

Exercises

1. An urn contains five red, three orange, and two blue balls. Two balls are randomly selected. What is the sample space of this experiment? Let X represent the number of orange balls selected. What are the possible values of X? Calculate $P\{X = 0\}$.

2. Let X represent the difference between the number of heads and the number of tails obtained when a coin is tossed n times. What are the possible values of X?

3. In Exercise 2, if the coin is assumed fair, then, for $n = 2$, what are the probabilities associated with the values that X can take on?

***4.** Suppose a die is rolled twice. What are the possible values that the following random variables can take on?

(a) The maximum value to appear in the two rolls.

(b) The minimum value to appear in the two rolls.

(c) The sum of the two rolls.

(d) The value of the first roll minus the value of the second roll.

5. If the die in Exercise 4 is assumed fair, calculate the probabilities associated with the random variables in (a)–(d).

6. Suppose five fair coins are tossed. Let E be the event that all coins land heads. Define the random variable I_E

$$I_E = \begin{cases} 1, & \text{if } E \text{ occurs} \\ 0, & \text{if } E^c \text{ occurs} \end{cases}$$

For what outcomes in the original sample space does I_E equal 1? What is $P\{I_E = 1\}$?

7. Suppose a coin having probability 0.7 of coming up heads is tossed three times. Let X denote the number of heads that appear in the three tosses. Determine the probability mass function of X.

8. Suppose the distribution function of X is given by

$$F(b) = \begin{cases} 0, & b < 0 \\ \frac{1}{2}, & 0 \le b < 1 \\ 1, & 1 \le b < \infty \end{cases}$$

What is the probability mass function of X?

9. If the distribution function of F is given by

$$F(b) = \begin{cases} 0, & b < 0 \\ \frac{1}{2}, & 0 \le b < 1 \\ \frac{3}{5}, & 1 \le b < 2 \\ \frac{4}{5}, & 2 \le b < 3 \\ \frac{9}{10}, & 3 \le b < 3.5 \\ 1, & b \ge 3.5 \end{cases}$$

calculate the probability mass function of X.

10. Suppose three fair dice are rolled. What is the probability at most one six appears?

***11.** A ball is drawn from an urn containing three white and three black balls. After the ball is drawn, it is then replaced and another ball is drawn. This goes on indefinitely. What is the probability that of the first four balls drawn, exactly two are white?

12. On a multiple-choice exam with three possible answers for each of the five questions, what is the probability that a student would get four or more correct answers just by guessing?

13. An individual claims to have extrasensory perception (ESP). As a test, a fair coin is flipped ten times, and he is asked to predict in advance the outcome. Our individual gets seven out of ten correct. What is the probability he would have done at least this well if he had no ESP? (Explain why the relevant probability is $P\{X \ge 7\}$ and not $P\{X = 7\}$.)

14. Suppose X has a binomial distribution with parameters 6 and $\frac{1}{2}$. Show that $X = 3$ is the most likely outcome.

15. Let X be binomially distributed with parameters n and p. Show that as k goes from 0 to n, $P(X = k)$ increases monotonically, then decreases monotonically, reaching its largest value

(a) in the case that $(n+1)p$ is an integer, when k equals either $(n+1)p - 1$ or $(n+1)p$,

(b) in the case that $(n+1)p$ is not an integer, when k satisfies $(n+1)p - 1 < k < (n+1)p$.

Hint: Consider $P\{X = k\}/P\{X = k-1\}$ and see for what values of k it is greater or less than 1.

***16.** An airline knows that 5 percent of the people making reservations on a certain flight will not show up. Consequently, their policy is to sell 52 tickets for a

flight that can hold only 50 passengers. What is the probability that there will be a seat available for every passenger who shows up?

17. Suppose that an experiment can result in one of r possible outcomes, the ith outcome having probability p_i, $i = 1, \ldots, r$, $\sum_{i=1}^{r} p_i = 1$. If n of these experiments are performed, and if the outcome of any one of the n does not affect the outcome of the other $n - 1$ experiments, then show that the probability that the first outcome appears x_1 times, the second x_2 times, and the rth x_r times is

$$\frac{n!}{x_1!x_2!\ldots x_r!}\, p_1^{x_1} p_2^{x_2} \cdots p_r^{x_r} \qquad \text{when } x_1 + x_2 + \cdots + x_r = n$$

This is known as the *multinomial* distribution.

18. In Exercise 17, let X_i denote the number of times that the ith type outcome occurs, $i = 1, \ldots, r$.

(a) For $0 \le j \le n$, use the definition of conditional probability to find $P(X_i = x_i, i = 1, \ldots, r-1 | X_r = j)$.

(b) What can you conclude about the conditional distribution of $X_1, \ldots, X_{r-1}$ given that $X_r = j$?

(c) Give an intuitive explanation for your answer to part (b).

19. In Exercise 17, let X_i denote the number of times the ith outcome appears, $i = 1, \ldots, r$. What is the probability mass function of $X_1 + X_2 + \ldots + X_k$?

20. In this problem we employ the multinomial distribution to solve an extension of the birthday problem. Assuming that each of n individuals is, independently of others, equally likely to have their birthday be any of the 365 days of the year, we want to derive an expression for the probability that the collection of n individuals will contain at least 3 having the same birthday.

(a) For a given partition of the 365 days of the year into a first set of size i, a second set of size $n - 2i$ and a third of size $365 - n + i$, find the probability that every day in the first set is the birthday of exactly 2 of the n individuals, every day in the second set is the birthday of exactly 1 of the n individuals, and every day in the third set is the birthday of none of the n individuals.

(b) For a given value i, determine the number of different partitions of the 365 days of the year into a first set of size i, a second set of size $n - 2i$ and a third set of size $365 - n + i$.

(c) Give an expression for the probability that a collection of n individuals does not contain at least 3 having the same birthday.

Remark. A computation gives that

$$1 - \sum_{i=0}^{44} \frac{365!}{i!(88-2i)!(365-88+i)!} \frac{88!}{2^i} \left(\frac{1}{365}\right)^{88} \approx .504.$$

21. Let X_1 and X_2 be independent binomial random variables, with X_i having parameters (n_i, p_i), $i = 1, 2$.

(a) Find $P(X_1 X_2 = 0)$.

(b) Find $P(X_1 + X_2 = 1)$.
(c) Find $P(X_1 + X_2 = 2)$.

22. If a fair coin is successively flipped, find the probability that a head first appears on the fifth trial.

***23.** A coin having probability p of coming up heads is successively flipped until the rth head appears. Argue that X, the number of flips required, will be $n, n \geq r$, with probability

$$P\{X = n\} = \binom{n-1}{r-1} p^r (1-p)^{n-r}, \qquad n \geq r$$

This is known as the negative binomial distribution.

Hint: How many successes must there be in the first $n-1$ trials?

24. The probability mass function of X is given by

$$p(k) = \binom{r+k-1}{r-1} p^r (1-p)^k, \qquad k = 0, 1, \ldots$$

Give a possible interpretation of the random variable X.

Hint: See Exercise 23.

In Exercises 25 and 26, suppose that two teams are playing a series of games, each of which is independently won by team A with probability p and by team B with probability $1 - p$. The winner of the series is the first team to win i games.

25. If $i = 4$, find the probability that a total of 7 games are played. Also show that this probability is maximized when $p = 1/2$.

26. Find the expected number of games that are played when
(a) $i = 2$;
(b) $i = 3$.
In both cases, show that this number is maximized when $p = 1/2$.

***27.** A fair coin is independently flipped n times, k times by A and $n - k$ times by B. Show that the probability that A and B flip the same number of heads is equal to the probability that there are a total of k heads.

28. Suppose that we want to generate a random variable X that is equally likely to be either 0 or 1, and that all we have at our disposal is a biased coin that, when flipped, lands on heads with some (unknown) probability p. Consider the following procedure:

1. Flip the coin, and let 0_1, either heads or tails, be the result.
2. Flip the coin again, and let 0_2 be the result.
3. If 0_1 and 0_2 are the same, return to step 1.
4. If 0_2 is heads, set $X = 0$, otherwise set $X = 1$.

(a) Show that the random variable X generated by this procedure is equally likely to be either 0 or 1.
(b) Could we use a simpler procedure that continues to flip the coin until the last two flips are different, and then sets $X = 0$ if the final flip is a head, and sets $X = 1$ if it is a tail?

29. Consider n independent flips of a coin having probability p of landing heads. Say a changeover occurs whenever an outcome differs from the one preceding it. For instance, if the results of the flips are $H\ H\ T\ H\ T\ H\ H\ T$, then there are a total of five changeovers. If $p = 1/2$, what is the probability there are k changeovers?

30. Let X be a Poisson random variable with parameter λ. Show that $P\{X = i\}$ increases monotonically and then decreases monotonically as i increases, reaching its maximum when i is the largest integer not exceeding λ.

Hint: Consider $P\{X = i\}/P\{X = i - 1\}$.

31. Compare the Poisson approximation with the correct binomial probability for the following cases:

(a) $P\{X = 2\}$ when $n = 8$, $p = 0.1$.
(b) $P\{X = 9\}$ when $n = 10$, $p = 0.95$.
(c) $P\{X = 0\}$ when $n = 10$, $p = 0.1$.
(d) $P\{X = 4\}$ when $n = 9$, $p = 0.2$.

32. If you buy a lottery ticket in 50 lotteries, in each of which your chance of winning a prize is $\frac{1}{100}$, what is the (approximate) probability that you will win a prize (a) at least once, (b) exactly once, (c) at least twice?

33. Let X be a random variable with probability density

$$f(x) = \begin{cases} c(1 - x^2), & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

(a) What is the value of c?
(b) What is the cumulative distribution function of X?

34. Let the probability density of X be given by

$$f(x) = \begin{cases} c(4x - 2x^2), & 0 < x < 2 \\ 0, & \text{otherwise} \end{cases}$$

(a) What is the value of c?
(b) $P\left\{\frac{1}{2} < X < \frac{3}{2}\right\} = ?$

35. The density of X is given by

$$f(x) = \begin{cases} 10/x^2, & \text{for } x > 10 \\ 0, & \text{for } x \le 10 \end{cases}$$

What is the distribution function of X? Find $P\{X > 20\}$.

36. A point is uniformly distributed within the disk of radius 1. That is, its density is

$$f(x, y) = C, \qquad 0 \le x^2 + y^2 \le 1$$

Find the probability that its distance from the origin is less than $x, 0 \le x \le 1$.

37. Let $X_1, X_2, \ldots, X_n$ be independent random variables, each having a uniform distribution over $(0, 1)$. Let $M = \text{maximum } (X_1, X_2, \ldots, X_n)$. Show that the distribution function of M, $F_M(\cdot)$, is given by

$$F_M(x) = x^n, \qquad 0 \le x \le 1$$

What is the probability density function of M?

38. Let $X_1, \ldots, X_{10}$ be independent and identically distributed continuous random variables with distribution function F, and mean $\mu = E[X_i]$. Let $X_{(1)} < X_{(2)} < \cdots < X_{(10)}$ be the values arranged in increasing order. That is, for $i = 1, \ldots, 10$, $X_{(i)}$ is the ith smallest of $X_1, \ldots, X_{10}$.

(a) Find $E[\sum_{i=1}^{10} X_{(i)}]$.

(b) Let $N = \max\{i : X_{(i)} < x\}$. What is the distribution of N.

(c) If m is the median of the distribution (that is, if $F(m) = .5$), find $P(X_{(2)} < m < X_{(8)})$.

39. An urn has 8 red and 12 blue balls. Suppose that balls are chosen at random and removed from the urn, with the process stopping when all the red balls have been removed. Let X be the number of balls that have been removed when the process stops.

(a) Find $P(X = 14)$.

(b) Find the probability that a specified blue ball remains in the urn.

(c) Find $E[X]$.

40. Suppose that two teams are playing a series of games, each of which is independently won by team A with probability p and by team B with probability $1 - p$. The winner of the series is the first team to win four games. Find the expected number of games that are played, and evaluate this quantity when $p = 1/2$.

41. Consider the case of arbitrary p in Exercise 29. Compute the expected number of changeovers.

42. Suppose that each coupon obtained is, independent of what has been previously obtained, equally likely to be any of m different types. Find the expected number of coupons one needs to obtain in order to have at least one of each type.

Hint: Let X be the number needed. It is useful to represent X by

$$X = \sum_{i=1}^{m} X_i$$

where each X_i is a geometric random variable.

43. An urn contains $n + m$ balls, of which n are red and m are black. They are withdrawn from the urn, one at a time and without replacement. Let X be the number of red balls removed before the first black ball is chosen. We are interested in determining $E[X]$. To obtain this quantity, number the red balls from 1 to n. Now define the random variables $X_i, i = 1, \ldots, n$, by

$$X_i = \begin{cases} 1, & \text{if red ball } i \text{ is taken before any black ball is chosen} \\ 0, & \text{otherwise} \end{cases}$$

(a) Express X in terms of the X_i.
(b) Find $E[X]$.

44. In Exercise 43, let Y denote the number of red balls chosen after the first but before the second black ball has been chosen.
(a) Express Y as the sum of n random variables, each of which is equal to either 0 or 1.
(b) Find $E[Y]$.
(c) Compare $E[Y]$ to $E[X]$ obtained in Exercise 43.
(d) Can you explain the result obtained in part (c)?

45. A total of r keys are to be put, one at a time, in k boxes, with each key independently being put in box i with probability $p_i, \sum_{i=1}^{k} p_i = 1$. Each time a key is put in a nonempty box, we say that a collision occurs. Find the expected number of collisions.

46. If X is a nonnegative integer valued random variable, show that

(a) $$E[X] = \sum_{n=1}^{\infty} P\{X \geq n\} = \sum_{n=0}^{\infty} P\{X > n\}$$

Hint: Define the sequence of random variables $I_n, n \geq 1$, by

$$I_n = \begin{cases} 1, & \text{if } n \leq X \\ 0, & \text{if } n > X \end{cases}$$

Now express X in terms of the I_n.

(b) If X and Y are both nonnegative integer valued random variables, show that

$$E[XY] = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} P(X \geq n, Y \geq m)$$

***47.** Consider three trials, each of which is either a success or not. Let X denote the number of successes. Suppose that $E[X] = 1.8$.
(a) What is the largest possible value of $P\{X = 3\}$?
(b) What is the smallest possible value of $P\{X = 3\}$?
In both cases, construct a probability scenario that results in $P\{X = 3\}$ having the desired value.

48. For any event A, we define the random variable $I\{A\}$, called the *indicator variable* for A, by letting it equal 1 when A occurs and 0 when A does not. Now, if $X(t)$ is a nonnegative random variable for all $t \geq 0$, then it follows from a result in real analysis called Fubini's theorem that

$$E\left[\int_0^{\infty} X(t)\, dt\right] = \int_0^{\infty} E[X(t)]\, dt$$

Suppose that X is a nonnegative random variable and that g is a differentiable function such that $g(0) = 0$.

(a) Show that

$$g(X) = \int_0^\infty I\{t < X\} g'(t)\, dt$$

(b) Show that

$$E[g(X)] = \int_0^\infty \bar{F}(t)\, g'(t)\, dt$$

where $\bar{F}(t) = 1 - F(t) = P(X > t)$.

***49.** Prove that $E[X^2] \geq (E[X])^2$. When do we have equality?

50. Let c be a constant. Show that

(a) $\text{Var}(cX) = c^2\text{Var}(X)$;

(b) $\text{Var}(c + X) = \text{Var}(X)$.

51. A coin, having probability p of landing heads, is flipped until a head appears for the rth time. Let N denote the number of flips required. Calculate $E[N]$.

Hint: There is an easy way of doing this. It involves writing N as the sum of r geometric random variables.

52. **(a)** Calculate $E[X]$ for the maximum random variable of Exercise 37.

(b) Calculate $E[X]$ for X as in Exercise 33.

(c) Calculate $E[X]$ for X as in Exercise 34.

53. If X is uniform over (0, 1), calculate $E[X^n]$ and $\text{Var}(X^n)$.

54. Each member of a population is either type 1 with probability p_1 or type 2 with probability $p_2 = 1 - p_1$. Independent of other pairs, two individuals of the same type will be friends with probability α, whereas two individuals of different types will be friends with probability β. Let P_i be the probability that a type i person will be friends with a randomly chosen other person.

(a) Find P_1 and P_2.

Let $F_{k,r}$ be the event that persons k and r are friends.

(b) Find $P(F_{1,2})$.

(c) Show that $P(F_{1,2}|F_{1,3}) \geq P(F_{1,2})$.

Hint for (c): It might be useful to let X be such that $P(X = P_i) = p_i, i = 1, 2$.

55. Suppose that the joint probability mass function of X and Y is

$$P(X = i, Y = j) = \binom{j}{i} e^{-2\lambda}\lambda^j/j!, \quad 0 \leq i \leq j$$

(a) Find the probability mass function of Y.

(b) Find the probability mass function of X.

(c) Find the probability mass function of $Y - X$.

56. There are n types of coupons. Each newly obtained coupon is, independently, type i with probability $p_i, i = 1, \ldots, n$. Find the expected number and the variance of the number of distinct types obtained in a collection of k coupons.

57. Suppose that X and Y are independent binomial random variables with parameters (n, p) and (m, p). Argue probabilistically (no computations necessary) that $X + Y$ is binomial with parameters $(n + m, p)$.

58. An urn contains $2n$ balls, of which r are red. The balls are randomly removed in n successive pairs. Let X denote the number of pairs in which both balls are red.

(a) Find $E[X]$.

(b) Find $\text{Var}(X)$.

59. Let X_1, X_2, X_3, and X_4 be independent continuous random variables with a common distribution function F and let

$$p = P\{X_1 < X_2 > X_3 < X_4\}$$

(a) Argue that the value of p is the same for all continuous distribution functions F.

(b) Find p by integrating the joint density function over the appropriate region.

(c) Find p by using the fact that all 4! possible orderings of $X_1, \ldots, X_4$ are equally likely.

60. Let X and Y be independent random variables with means μ_x and μ_y and variances σ_x^2 and σ_y^2. Show that

$$\text{Var}(XY) = \sigma_x^2\sigma_y^2 + \mu_y^2\sigma_x^2 + \mu_x^2\sigma_y^2$$

61. Let $X_1, X_2, \ldots$ be a sequence of independent identically distributed continuous random variables. We say that a record occurs at time n if $X_n > \max(X_1, \ldots, X_{n-1})$. That is, X_n is a record if it is larger than each of $X_1, \ldots, X_{n-1}$. Show

(a) $P\{\text{a record occurs at time } n\} = 1/n$;

(b) $E[\text{number of records by time } n] = \sum_{i=1}^{n} 1/i$;

(c) $\text{Var}(\text{number of records by time } n) = \sum_{i=1}^{n}(i-1)/i^2$;

(d) Let $N = \min\{n: n > 1 \text{ and a record occurs at time } n\}$. Show $E[N] = \infty$.

Hint: For (b) and (c) represent the number of records as the sum of indicator (that is, Bernoulli) random variables.

62. Let $a_1 < a_2 < \cdots < a_n$ denote a set of n numbers, and consider any permutation of these numbers. We say that there is an inversion of a_i and a_j in the permutation if $i < j$ and a_j precedes a_i. For instance the permutation 4, 2, 1, 5, 3 has 5 inversions—(4, 2), (4, 1), (4, 3), (2, 1), (5, 3). Consider now a random permutation of $a_1, a_2, \ldots, a_n$—in the sense that each of the $n!$ permutations is equally likely to be chosen—and let N denote the number of inversions in this permutation. Also, let

$$N_i = \text{number of } k: k < i,\ a_i \text{ precedes } a_k \text{ in the permutation}$$

and note that $N = \sum_{i=1}^{n} N_i$.

(a) Show that $N_1, \ldots, N_n$ are independent random variables.

(b) What is the distribution of N_i?
(c) Compute $E[N]$ and $\text{Var}(N)$.

63. Let X denote the number of white balls selected when k balls are chosen at random from an urn containing n white and m black balls.
(a) Compute $P\{X = i\}$.
(b) Let, for $i = 1, 2, \ldots, k$; $j = 1, 2, \ldots, n$,

$$X_i = \begin{cases} 1, & \text{if the } i\text{th ball selected is white} \\ 0, & \text{otherwise} \end{cases}$$

$$Y_j = \begin{cases} 1, & \text{if white ball } j \text{ is selected} \\ 0, & \text{otherwise} \end{cases}$$

Compute $E[X]$ in two ways by expressing X first as a function of the X_is and then of the Y_js.

***64.** Show that $\text{Var}(X) = 1$ when X is the number of men who select their own hats in Example 2.30.

65. The number of traffic accidents on successive days are independent Poisson random variables with mean 2.
(a) Find the probability that 3 of the next 5 days have two accidents.
(b) Find the probability that there are a total of six accidents over the next 2 days.
(c) If each accident is independently a "major accident" with probability p, what is the probability there are no major accidents tomorrow?

***66.** Show that the random variables $X_1, \ldots, X_n$ are independent if for each $i = 2, \ldots, n$, X_i is independent of $X_1, \ldots, X_{i-1}$.

Hint: $X_1, \ldots, X_n$ are independent if for any sets $A_1, \ldots, A_n$

$$P(X_j \in A_j, j = 1, \ldots, n) = \prod_{j=1}^{n} P(X_j \in A_j)$$

On the other hand X_i is independent of $X_1, \ldots, X_{i-1}$ if for any sets $A_1, \ldots, A_i$

$$P(X_i \in A_i | X_j \in A_j, j = 1, \ldots, i-1) = P(X_i \in A_i)$$

67. Calculate the moment generating function of the uniform distribution on $(0, 1)$. Obtain $E[X]$ and $\text{Var}[X]$ by differentiating.

68. Let X and W be the working and subsequent repair times of a certain machine. Let $Y = X + W$ and suppose that the joint probability density of X and Y is

$$f_{X,Y}(x, y) = \lambda^2 e^{-\lambda y}, \quad 0 < x < y < \infty$$

(a) Find the density of X.
(b) Find the density of Y.
(c) Find the joint density of X and W.
(d) Find the density of W.

69. In deciding upon the appropriate premium to charge, insurance companies sometimes use the exponential principle, defined as follows. With X as the random amount that it will have to pay in claims, the premium charged by the insurance company is

$$P = \frac{1}{a} \ln(E[e^{aX}])$$

where a is some specified positive constant. Find P when X is an exponential random variable with parameter λ, and $a = \alpha\lambda$, where $0 < \alpha < 1$.

70. Calculate the moment generating function of a geometric random variable.

***71.** Show that the sum of independent identically distributed exponential random variables has a gamma distribution.

72. Successive monthly sales are independent normal random variables with mean 100 and variance 100.

(a) Find the probability that at least one of the next 5 months has sales above 115.

(b) Find the probability that the total number of sales over the next 5 months exceeds 530.

73. Consider n people and suppose that each of them has a birthday that is equally likely to be any of the 365 days of the year. Furthermore, assume that their birthdays are independent, and let A be the event that no two of them share the same birthday. Define a "trial" for each of the $\binom{n}{2}$ pairs of people and say that trial (i, j), $i \neq j$, is a success if persons i and j have the same birthday. Let $S_{i,j}$ be the event that trial (i, j) is a success.

(a) Find $P(S_{i,j})$, $i \neq j$.

(b) Are $S_{i,j}$ and $S_{k,r}$ independent when i, j, k, r are all distinct?

(c) Are $S_{i,j}$ and $S_{k,j}$ independent when i, j, k are all distinct?

(d) Are $S_{1,2}, S_{1,3}, S_{2,3}$independent?

(e) Employ the Poisson paradigm to approximate $P(A)$.

(f) Show that this approximation yields that $P(A) \approx .5$ when $n = 23$.

(g) Let B be the event that no three people have the same birthday. Approximate the value of n that makes $P(B) \approx .5$. (Whereas a simple combinatorial argument explicitly determines $P(A)$, the exact determination of $P(B)$ is very complicated.)

Hint: Define a trial for each triplet of people.

***74.** If X is Poisson with parameter λ, show that its Laplace transform is given by

$$g(u) = E[e^{-uX}] = e^{\lambda(e^{-u}-1)}$$

75. Consider Example 2.48. Find $\text{Cov}(X_i, X_j)$ in terms of the a_{rs}.

76. Use Chebyshev's inequality to prove the *weak law of large numbers*. Namely, if $X_1, X_2, \ldots$ are independent and identically distributed with mean μ and vari-

ance σ^2 then, for any $\varepsilon > 0$,

$$P\left\{\left|\frac{X_1 + X_2 + \cdots + X_n}{n} - \mu\right| > \varepsilon\right\} \to 0 \qquad \text{as } n \to \infty$$

77. Suppose that X is a random variable with mean 10 and variance 15. What can we say about $P\{5 < X < 15\}$?

78. Let $X_1, X_2, \ldots, X_{10}$ be independent Poisson random variables with mean 1.
(a) Use the Markov inequality to get a bound on $P\{X_1 + \cdots + X_{10} \geq 15\}$.
(b) Use the central limit theorem to approximate $P\{X_1 + \cdots + X_{10} \geq 15\}$.

79. If X is normally distributed with mean 1 and variance 4, use the tables to find $P\{2 < X < 3\}$.

***80.** Show that

$$\lim_{n\to\infty} e^{-n} \sum_{k=0}^{n} \frac{n^k}{k!} = \frac{1}{2}$$

Hint: Let X_n be Poisson with mean n. Use the central limit theorem to show that $P\{X_n \leq n\} \to \frac{1}{2}$.

81. Let X and Y be independent normal random variables, each having parameters μ and σ^2. Show that $X + Y$ is independent of $X - Y$.

Hint: Find their joint moment generating function.

82. Let $\phi(t_1, \ldots, t_n)$ denote the joint moment generating function of $X_1, \ldots, X_n$.
(a) Explain how the moment generating function of X_i, $\phi_{X_i}(t_i)$, can be obtained from $\phi(t_1, \ldots, t_n)$.
(b) Show that $X_1, \ldots, X_n$ are independent if and only if

$$\phi(t_1, \ldots, t_n) = \phi_{X_1}(t_1) \cdots \phi_{X_n}(t_n)$$

83. With $K(t) = \log(E\left[e^{tX}\right])$, show that

$$K'(0) = E[X], \quad K''(0) = \text{Var}(X)$$

84. Teams 1, 2, 3, 4 are all scheduled to play each of the other teams 10 times. Whenever team i plays team j, team i is the winner with probability $P_{i,j}$, where

$$P_{1,2} = .6, \quad P_{1,3} = .7, \quad P_{1,4} = .75$$
$$P_{2,3} = .6, \quad P_{2,4} = .70, \quad P_{3,4} = .5$$

(a) Approximate the probability that team 1 wins at least 20 games.
Suppose now that we want to approximate the probability that team 2 wins at least as many games as does team 1. To do so, let X be the number of games that team 2 wins against team 1, let Y be the total number of games that team 2 wins against teams 3 and 4, and let Z be the total number of games that team 1 wins against teams 3 and 4.
(b) Are X, Y, Z independent.

(c) Express the event that team 2 wins at least as many games as does team 1 in terms of the random variables X, Y, Z.

(d) Approximate the probability that team 2 wins at least as many games as does team 1.

85. The *standard deviation* of a random variable is the positive square root of its variance. Letting σ_X and σ_Y denote the standard deviations of the random variables X and Y, we define the *correlation* of X and Y by

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

(a) Starting with the inequality $\text{Var}\left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}\right) \geq 0$, show that $-1 \leq \text{Corr}(X, Y)$.

(b) Prove the inequality

$$-1 \leq \text{Corr}(X, Y) \leq 1$$

(c) If σ_{X+Y} is the standard deviation of $X + Y$, show that

$$\sigma_{X+Y} \leq \sigma_X + \sigma_Y$$

***86.** Each new book donated to a library must be processed. Suppose that the time it takes a librarian to process a book has mean 10 minutes and standard deviation 3 minutes. If a librarian has 40 books that must be processed one at a time,

(a) approximate the probability that it will take more than 420 minutes to process all these books;

(b) approximate the probability that at least 25 books will be processed in the first 240 minutes.

87. Recall that X is said to be a gamma random variable with parameters (α, λ) if its density is

$$f(x) = \lambda e^{-\lambda x} (\lambda x)^{\alpha - 1} / \Gamma(\alpha), \quad x > 0$$

(a) If Z is a standard normal random variable, show that Z^2 is a gamma random variable with parameters $(1/2, 1/2)$.

(b) If $Z_1, \ldots, Z_n$ are independent standard normal random variables, then $\sum_{i=1}^{n} Z_i^2$ is said to be a *chi-squared* random variable with n degrees of freedom. Explain how you can use results from Example 2.39 to show that the density function of $\sum_{i=1}^{n} Z_i^2$ is

$$f(x) = \frac{e^{-x/2} x^{n/2-1}}{2^{n/2} \Gamma(n/2)}, \quad x > 0$$

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